








Modelling Financial Markets

Chapter 6: Multivariate Volatility and Dependence



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When Do Markets Move Together, and How Sure Can We Be?

What we will explain

- Part I: why dependence matters, covariance uncertainty, multivariate GARCH (generalised autoregressive conditional heteroskedasticity) and DCC (dynamic conditional correlation), crisis correlations and the Forbes–Rigobon bias
- Part II: copulas and tail dependence, European and Romanian markets, banks, common volatility and research with artificial intelligence (AI)
- Builds on Chapter 1 (fat tails, volatility clustering), Chapter 4 (Σ in portfolio choice) and Chapter 5 (univariate GARCH)
- Leads to Chapter 7 (portfolio Value at Risk, VaR, and Expected Shortfall, ES), Chapter 8 (backtesting) and Chapter 18 (systemic risk)

What you should be able to do

- estimate a time-varying covariance (rolling window, EWMA: exponentially weighted moving average, DCC) and attach a valid standard error to a correlation
- separate contagion from the volatility bias of crisis correlations
- choose, fit and test a copula, and read its tail dependence at a finite threshold and in the limit

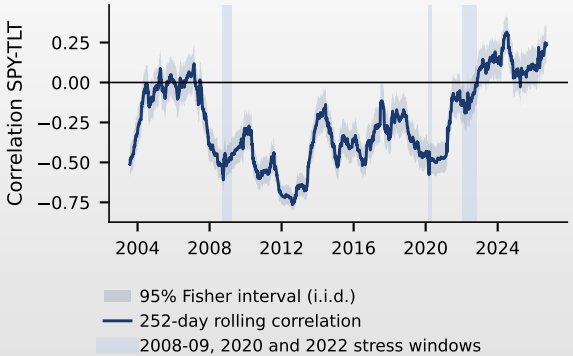
Why Dependence Matters (1/2): Portfolio Diversification

- **Diversification** (Markowitz (1952))
  - The portfolio variance depends on the correlation through a single cross term
  - $\sigma_p^2 = w^\top \Sigma w = w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\rho\,\sigma_1\sigma_2$
  - w: weight of asset 1 ($1-w$ in asset 2)
  - Σ: covariance matrix
  - σ_1, σ_2: volatilities
  - ρ: correlation
  - Portfolio 60/40 of SPY (S&P 500 exchange-traded fund) and TLT (20+ year US Treasury bond fund)
  - Annualised volatilities over July 2002 – September 2026: $\sigma_1 = 18.8\%$ (SPY), $\sigma_2 = 14.3\%$ (TLT); weights rebalanced daily to 60/40
  - We split at January 2022, the start of the inflation shock; the break test of the DCC section dates the change earlier, in August 2020
  - Portfolio volatility 10.4% with the 2002–2021 correlation (-0.40), 13.2% with the 2022–2026 one (0.11)

Why Dependence Matters (2/2): Hedges, Contagion and Moving Covariances

- **Hedging**
  - How much value of the hedge F to sell per unit of value held in the position S to minimise the variance
  - Minimum-variance hedge ratio $h^* = \rho\,\sigma_S/\sigma_F$: the slope of the regression of r_S on r_F. Estimated on returns, it is a ratio of values: units of F per unit of S are $h^* P_S/P_F$ (P_S, P_F: prices, contract size included)
  - r_S, r_F: returns of the position and of the hedge
  - σ_S, σ_F: their volatilities
  - Effectiveness ρ^2: the share of the variance removed
  - If the correlation moves, yesterday's hedge is wrong today: a dynamic hedge needs a dynamic ρ_t
- **Contagion and interdependence** (Forbes & Rigobon (2002))
  - Contagion: a *significant increase* in cross-market linkages after a shock
  - Interdependence: links present in all states (trade, common investors, common macro factors)
  - Correlations tend to rise exactly when diversification is needed most
- New in this chapter: Σ_t, the covariance matrix at date t, moves

Stocks and Bonds: A Correlation That Changed Sign



- Daily SPY and TLT returns, 252-day window
- Band: 95% Fisher interval assuming i.i.d. (independent and identically distributed) data
- Shaded: autumn 2008 – March 2009, February–April 2020, January–October 2022
- Minimum −0.77 (8 August 2012), maximum 0.32 (1 July 2024), last 0.24 (18 September 2026)
- Positive on 6% of days before 2022, on 81% since January 2022
- Growth shocks move stocks and bond prices in opposite directions (negative correlation)
- Inflation shocks hit both (positive, as in 2022)
- The Fisher band is too narrow here: it ignores fat tails and volatility clustering (next section)

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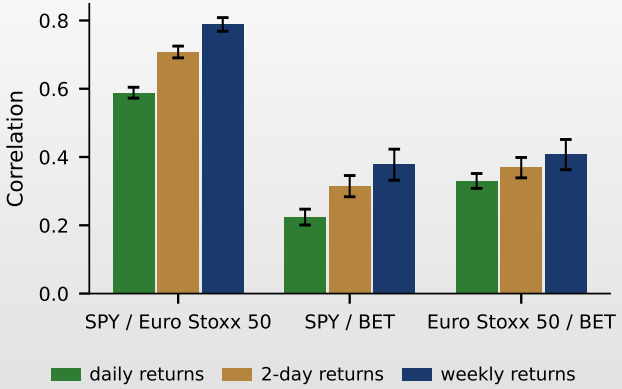
How Precise Is an Estimated Correlation?

- The question of this section: how do we estimate Σ, and how uncertain is each estimated correlation?
- First trap: closing times differ, so the data frequency changes the correlation
- Second trap: fat tails and volatility clustering make the Fisher interval too narrow
- Third trap: the window length decides how fast we see a regime change

Asynchronous Trading: A Trap for Daily Correlations

- Closing times differ: the Bucharest Stock Exchange and the euro-area exchanges close hours before New York
  - US news after the European close enters European prices only the next day
- Consequence (positive dependence, as between equity markets): the same-day correlation is biased towards zero, and the lagged correlation is positive
- Remedies
  - Lower frequency: 2-day or weekly returns on common trading days
  - Add the lead–lag term: $\rho_0 + \rho_{+1}$ (ρ_0: same-day correlation, ρ_{+1}: US yesterday with Europe today)
  - A rough approximation of the synchronous correlation, valid only for a one-day lag, no autocorrelation and equal variances
- Rule used throughout: first align the *prices* on common trading days, then compute returns

Daily, 2-Day and Weekly Correlations, 2000–2026 (1/2)



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Daily, 2-Day and Weekly Correlations, 2000–2026 (2/2)

- SPY, Euro Stoxx 50 and BET (the main index of the Bucharest Stock Exchange) on common trading days
- Vertical lines on the bars: 95% Fisher intervals
- SPY–Euro Stoxx 50: 0.59 daily, 0.71 on 2-day returns, 0.79 weekly
- SPY yesterday with Euro Stoxx 50 today: 0.20
- SPY–BET: 0.22 daily versus 0.38 weekly (lagged 0.17)
- Euro Stoxx 50–BET, almost synchronous closes: 0.33 versus 0.41
- Weekly data are cleaner but give 1 367 instead of 6 446 observations: wider intervals

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Three Frequencies, Each Chosen for a Reason

- Returns are log returns, $r_t = \ln P_t - \ln P_{t-1}$, computed after aligning the prices on common trading days
- Daily: DCC, HAC intervals (standard errors robust to heteroskedasticity and autocorrelation, defined below) and hedging, where the reaction speed matters
- 2-day: the Forbes–Rigobon tests, to absorb different closing times
- 2-day returns overlap in the Forbes–Rigobon tests (two consecutive returns share one day), but not in the three-frequency chart
- Weekly (Friday closes): exceedance correlations, market integration and bank networks, which mix time zones

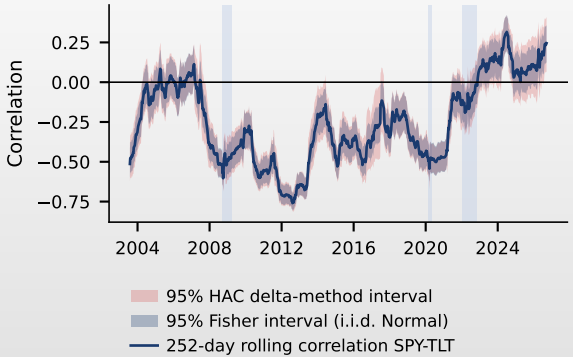
Sample Covariance and Inference on a Correlation (1/2): Estimator and Fisher Interval

- Sample covariance matrix and Pearson correlation of N assets over T days
  - $\hat{\Sigma} = \frac{1}{T-1} \sum_t (r_t - \bar{r})(r_t - \bar{r})^\top, \quad \hat{\rho}_{ij} = \hat{\Sigma}_{ij} / \sqrt{\hat{\Sigma}_{ii} \hat{\Sigma}_{jj}}$
  - $r_t \in \mathbb{R}^N$: vector of the N returns on day t, $t = 1, \ldots, T$
  - $\bar{r}$: vector of sample means
  - $\hat{\Sigma}_{ij}$: element (i, j)
  - $N(N+1)/2$ distinct elements to estimate: 125 250 for $N = 500$
- **Fisher transform**: $z = \operatorname{artanh} \hat{\rho} = \frac{1}{2} \ln \frac{1+\hat{\rho}}{1-\hat{\rho}}$ is approximately Normal with a variance that does not depend on ρ
  - $z \approx N\big(\operatorname{artanh} \rho,\ 1/(T-3)\big)$ for i.i.d. pairs with the bivariate Normal distribution
  - 95% interval $\tanh(z \pm 1.96/\sqrt{T-3})$
  - $T = 252$, $\rho = 0.5$: half-width about 0.09

Sample Covariance and Inference on a Correlation (2/2): Delta Method and HAC

- The delta method writes the estimation error $\hat{\rho} - \rho$ as a sample mean of daily contributions ψ_t
- Volatility clustering does not change this form, only the variance of ψ_t
  - $\hat{\rho} - \rho \approx \frac{1}{T} \sum_t \psi_t, \quad \psi_t = \tilde{x}_t \tilde{y}_t - \frac{\rho}{2}(\tilde{x}_t^2 + \tilde{y}_t^2)$
  - $\tilde{x}_t$, $\tilde{y}_t$: the two returns standardised to mean 0 and variance 1
  - ψ_t: the contribution of day t to the error
- Standard error (SE) from the long-run variance of ψ_t
  - HAC (heteroskedasticity and autocorrelation consistent): Bartlett kernel, L lags (Newey & West (1987))
  - Needs stationarity, finite fourth moments and weak dependence
  - Lags by the usual rule of thumb $L = \lfloor 4(T/100)^{2/9} \rfloor$: $L = 9$ for $T = 4\,891$, $L = 6$ for $T = 1\,182$, $L = 4$ for a 252-day window
  - i.i.d. Normal pairs give $\mathrm{Var}(\psi_t) = (1 - \rho^2)^2$, the Fisher variance
  - Fat tails and GARCH raise it (derivation in the Appendix)
- SPY–TLT daily, 2002–2021 ($T = 4\,891$, $\hat{\rho} = -0.403$): SE 0.0120 (Fisher), 0.0217 (fourth moments, $L = 0$), 0.0227 (HAC, $L = 9$)

Fisher versus HAC Intervals: SPY and TLT



- 252-day rolling correlation, every 5th day, with the 95% Fisher interval and the 95% HAC delta-method interval (Bartlett, $L = 4$)
- HAC is wider in almost every window: median width ratio 1.24, 90% quantile 1.54
- Correlation significantly different from zero: 73% of windows by Fisher, 65% by HAC
- The verdict flips in 8.2%
- Whole periods: HAC SE $= 1.9\times$ Fisher in 2002–2021, $1.4\times$ since 2022
- Fourth moments alone give $1.8\times$ in 2002–2021 ($1.28\times$ since 2022): fat tails matter more than autocorrelation
- Rule for daily returns: report delta-method HAC intervals

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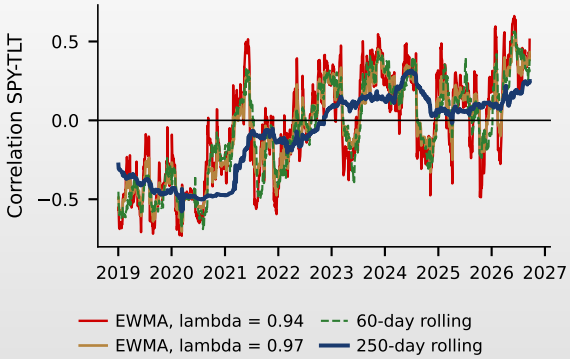
Rolling Windows

- **Rolling window** of length m: $\hat{\Sigma}_t = \frac{1}{m} \sum_{s=t-m}^{t-1} r_s r_s^\top$ (mean set to zero for daily data)
- r_s: return vector of day s; m: number of past days used
- Positive semidefinite: for any weight vector v, $v^\top \hat{\Sigma}_t v = \frac{1}{m} \sum_s (v^\top r_s)^2 \geq 0$, so every portfolio variance is ≥ 0
- Singular if $m < N$; positive definite only if the m return vectors span $\mathbb{R}^N$
- **Ghost effect**: an extreme day enters with weight $1/m$ and leaves abruptly m days later
- The trade-off: a short window reacts fast but is noisy; a long one is stable but detects a regime change late

EWMA (RiskMetrics)

- **EWMA** (exponentially weighted moving average, J.P. Morgan/Reuters (1996)): $\boldsymbol{\Sigma}_t = \lambda \boldsymbol{\Sigma}_{t-\mathbf{1}} + (1-\lambda)\, r_{t-\mathbf{1}} r_{t-\mathbf{1}}^\top = (1-\lambda) \sum_{k \geq \mathbf{1}} \lambda^{k-\mathbf{1}} r_{t-k} r_{t-k}^\top$
- $\lambda \in (0,1)$: decay factor; k: age of the observation in periods, weight $(1-\lambda)\lambda^{k-\mathbf{1}}$
- Usual values: $\lambda = 0.94$ for daily and $\lambda = 0.97$ for monthly data
- Half-life $\ln 0.5 / \ln \lambda$: 11.2 periods for $\lambda = 0.94$ and 22.8 for $\lambda = 0.97$; effective sample $(1+\lambda)/(1-\lambda)$ of about 32 days for $\lambda = 0.94$
- Positive semidefinite, no ghost effect
- Limitation: one λ for all elements and no mean reversion, i.e. an integrated GARCH(1,1) without a constant (Chapter 5)
- λ plays the role of m: the same trade-off between reaction speed and noise

EWMA versus Rolling Windows, SPY–TLT since 2019



- Two EWMA estimators ($\lambda = 0.94$ and 0.97) and two rolling windows (60 and 250 days) on the same data
- Standard deviation of the daily change of the estimate: 0.060 (EWMA 0.94), 0.032 (EWMA 0.97), 0.0088 (250-day window)
- From 10 May 2022 the EWMA correlation with $\lambda = 0.94$ stays positive for 20 consecutive days
- The 250-day window gives the same signal only on 8 December 2022
- With EWMA and rolling windows, λ and m are chosen by the analyst, not estimated
- DCC (next section) estimates the reaction speed and the memory of the correlation from the data

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From Univariate to Multivariate Volatility (1/2): the Model

- The question of this section: how do we model a covariance matrix that changes every day, stays positive definite and has few parameters?
- Chapter 5: one asset, $r_t = \sigma_t z_t$, $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$; now N assets and a covariance matrix
  - $r_t = H_t^{1/2} z_t, \quad \mathbb{E}(z_t \mid \mathcal{F}_{t-1}) = 0, \quad \mathrm{Var}(z_t \mid \mathcal{F}_{t-1}) = I_N, \quad$ so $\mathrm{Var}(r_t \mid \mathcal{F}_{t-1}) = H_t$
  - r_t, z_t: vectors of N returns and innovations
  - H_t: $N \times N$ conditional covariance matrix, known at $t-1$ (Σ_t in the previous section)
  - $H_t^{1/2}$: a matrix square root, $H_t^{1/2}(H_t^{1/2})^\top = H_t$
  - I_N: identity matrix
  - $\mathcal{F}_{t-1}$: information up to $t-1$
- Any model for H_t must deliver
  - positive definiteness at every t (every portfolio has a positive variance)
  - Few parameters
  - Interpretable dynamics
- Gaussian quasi-log-likelihood, maximised over the parameters of H_t
  - $\ell = -\frac{1}{2} \sum_t \left(N \ln 2\pi + \ln |H_t| + r_t^\top H_t^{-1} r_t \right)$
  - $|H_t|$: determinant of H_t
  - Quasi: the estimates stay consistent if z_t is not Normal but H_t is right

From Univariate to Multivariate Volatility (2/2): the Curse of Dimension

- **VEC** (Bollerslev, Engle & Wooldridge (1988)):
  $\operatorname{vech}(H_t) = c + A \operatorname{vech}(r_{t-\mathbf{1}} r_{t-\mathbf{1}}^\top) + B \operatorname{vech}(H_{t-\mathbf{1}})$
  - vech: the vector of the $N(N+1)/2$ distinct elements of a symmetric matrix; A, B: square matrices of that size
  - Positive definiteness not guaranteed
- **BEKK** (Baba–Engle–Kraft–Kroner, Engle & Kroner (1995)):
  $H_t = CC^\top + A^\top r_{t-\mathbf{1}} r_{t-\mathbf{1}}^\top A + B^\top H_{t-\mathbf{1}} B$
  - C: lower triangular; A, B: $N \times N$ matrices
  - Positive definite by construction (a sum of quadratic forms), but the elements of A, B have no direct interpretation (▸ Appendix)
- **DCC**: $3N$ univariate GARCH parameters plus a, b; the target $\bar{Q}$ is estimated separately (next slide)
- The number of parameters grows like N^4 (VEC) and N^2 (BEKK): infeasible for large portfolios

Number of parameters

N	VEC	full BEKK	DCC
2	21	11	$6+2$
5	465	65	$15+2$
10	6 105	255	$30+2$
50	3 252 525	6 275	$150+2$

Constant and Dynamic Conditional Correlation

- Decomposition: covariance = volatilities $\times$ correlations $\times$ volatilities
  - $H_t = D_t R_t D_t, \quad D_t = \text{diag}(\sigma_{\mathbf{1}t}, \ldots, \sigma_{Nt})$
  - σ_{it}: GARCH volatility of asset i; R_t: conditional correlation matrix; $\varepsilon_{it} = r_{it}/\sigma_{it}$: standardised residuals
- **CCC** (constant conditional correlation, Bollerslev (1990)): $R_t = R$ constant
- **DCC** (Engle (2002)): a GARCH-type recursion for a proxy matrix Q_t
  - $Q_t = (1 - a - b)\,\bar{Q} + a\,\varepsilon_{t-\mathbf{1}}\varepsilon_{t-\mathbf{1}}^\top + b\,Q_{t-\mathbf{1}}, \quad a, b \geq 0,\ a + b < 1$
  - $R_t = \text{diag}(Q_t)^{-\mathbf{1/2}}\, Q_t\, \text{diag}(Q_t)^{-\mathbf{1/2}}$
  - $\bar{Q}$: the intercept matrix of the recursion, with $N(N-1)/2$ free elements. Correlation targeting replaces it by the sample covariance $\hat{\bar{Q}} = \frac{\mathbf{1}}{T} \sum_t \varepsilon_t \varepsilon_t^\top$ (when this is consistent: slide "Two-Step Estimation (2/2)")
  - a: reaction to yesterday's product of shocks; b: memory; $a + b$: persistence
  - The second line rescales Q_t so that its diagonal is 1: R_t is a valid correlation matrix
- Two parameters govern all $N(N-1)/2$ correlations: parsimonious, but every pair has the same persistence
- Q_t is a covariance proxy, not a correlation: $\mathbb{E}_{t-\mathbf{1}}[\varepsilon_t \varepsilon_t^\top] = R_t \neq Q_t$

Two-Step Estimation (1/2): Splitting the Likelihood

- The Gaussian log-likelihood splits into a volatility part and a correlation part
  - $\ell = \underbrace{-\tfrac{1}{2}\sum_t \left(N \ln 2\pi + 2\ln|D_t| + r_t^\top D_t^{-2} r_t\right)}_{\ell_V} \underbrace{-\tfrac{1}{2}\sum_t \left(\ln|R_t| + \varepsilon_t^\top R_t^{-1}\varepsilon_t - \varepsilon_t^\top \varepsilon_t\right)}_{\ell_C}$
  - ℓ_V: depends only on the GARCH parameters (volatility part)
  - ℓ_C: depends on (a, b) given the volatilities (correlation part)
  - $|D_t|$, $|R_t|$: determinants
- Where the split comes from: substitute $H_t = D_t R_t D_t$ and $\varepsilon_t = D_t^{-1} r_t$ into the quasi-log-likelihood of the previous section
  - Determinant: $|H_t| = |D_t|^2 |R_t|$, so $\ln|H_t| = 2\ln|D_t| + \ln|R_t|$
  - Quadratic form: $r_t^\top H_t^{-1} r_t = r_t^\top D_t^{-1} R_t^{-1} D_t^{-1} r_t = \varepsilon_t^\top R_t^{-1} \varepsilon_t$
  - Add and subtract $\varepsilon_t^\top \varepsilon_t = r_t^\top D_t^{-2} r_t = \sum_i r_{it}^2/\sigma_{it}^2$: then $\ell_V = \sum_i \left[-\tfrac{1}{2}\sum_t (\ln 2\pi + \ln\sigma_{it}^2 + r_{it}^2/\sigma_{it}^2)\right]$, the sum of the N univariate Gaussian GARCH log-likelihoods
- Step 1: fit N univariate GARCH(1,1) models, keep σ_{it} and ε_{it}
- Step 2: fix $\bar{Q}$ at the sample covariance of ε_t and maximise ℓ_C over (a, b)

Two-Step Estimation (2/2): Targeting and Inference

- Targeting with the sample covariance makes the standard DCC estimator inconsistent in general. Why: take expectations of the recursion, using $\mathbb{E}[\varepsilon_{t-\mathbf{1}}\varepsilon_{t-\mathbf{1}}^\top] = \mathbb{E}[R_{t-\mathbf{1}}]$
  - Stationarity gives $(1-b)\,\mathbb{E}[Q_t] = (1-a-b)\,\bar{Q} + a\,\mathbb{E}[R_t]$
  - $\hat{\bar{Q}} = \frac{\mathbf{1}}{T}\sum_t \varepsilon_t\varepsilon_t^\top$ estimates $\mathbb{E}[R_t]$, which equals the intercept $\bar{Q}$ only if $\mathbb{E}[Q_t] = \mathbb{E}[R_t]$
  - That fails in general: R_t rescales Q_t by its own random diagonal
  - The corrected DCC of Aielli (2013) uses $\mathrm{diag}(Q_{t-\mathbf{1}})^{\mathbf{1}/\mathbf{2}}\varepsilon_{t-\mathbf{1}}$ in place of $\varepsilon_{t-\mathbf{1}}$, so that $\mathbb{E}[Q_t] = \bar{Q}$ and targeting remains consistent
- Step-2 standard errors treat the GARCH estimates and $\bar{Q}$ as known: propagate their error with the stacked sandwich of Newey & McFadden (1994) or by bootstrapping both steps (▸ Appendix)
- CCC against DCC: $H_{\mathbf{0}} : a = 0$
- b is not identified under $H_{\mathbf{0}}$, so the $\chi^{\mathbf{2}}$ reference is only indicative: bootstrap under CCC

DCC Estimates for SPY and TLT

Step 1: GARCH(1,1), returns in %

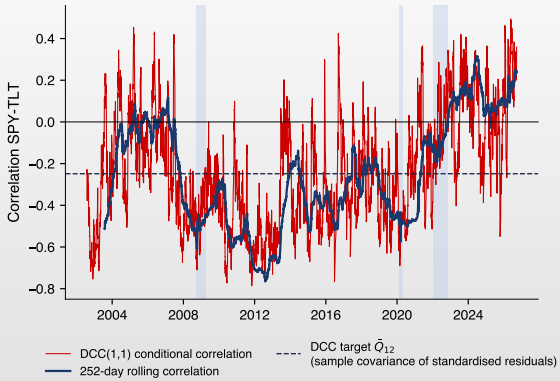
	ω	α	β	$\alpha+\beta$
SPY	0.0283	0.127	0.848	0.975
TLT	0.0057	0.049	0.943	0.993

Step 2: DCC(1,1) and three standard errors

	estimate	Hessian	boot. step 2	boot. both
a	0.0522	0.0059	0.0051	0.0052
b	0.9335	0.0084	0.0077	0.0077
$a+b$	0.9857	–	0.0038	0.0040

- Daily data 31 July 2002 – 18 September 2026, $T = 6\,073$; target $\bar{\tilde{Q}}_{\mathbf{12}} = -0.248$
- Persistence $a + b = 0.986$: a shock to Q_t halves in $\ln 0.5 / \ln 0.986 \approx 48$ days (only approximately for R_t, since the recursion is linear in Q_t)
- LR test against CCC: $LR = 2(\hat{\ell}^{DCC} - \hat{\ell}^{CCC}) = 2(460.26 - 193.54) = 533.4$
- Bootstrap of both steps under CCC ($B = 199$): the largest simulated value under H_0 is 15.7, so $p = 1/(B+1) = 0.005$, the smallest attainable value: CCC is rejected
- Step-1 uncertainty raises the SE by only 2% for a and 1% for b, because T is large
- The Hessian SEs are larger than the bootstrap ones, hence conservative
- TLT volatility is more persistent than SPY's but reacts less to news (smaller α)

DCC Correlation of SPY and TLT, 2002–2026



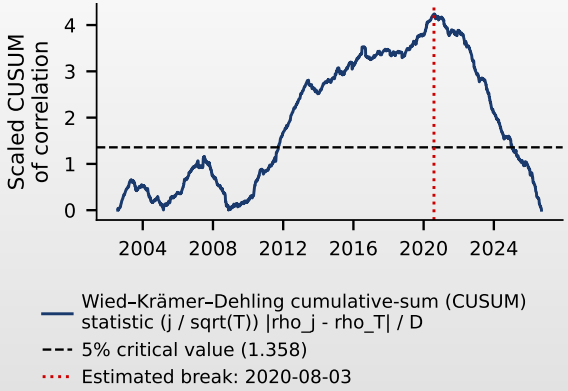
- DCC(1,1) conditional correlation, the 252-day rolling correlation and the target $\bar{\hat{Q}}_{12} = -0.248$
- Range -0.79 (2 November 2011) to 0.49 (21 May 2026)
- Last value 0.36
- Average: -0.54 from 15 September to 31 December 2008, -0.56 in 2012, 0.00 in 2022
- Since January 2022 the path is above the target on 92% of days (36% before): a single long-run correlation is doubtful

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Testing Constant Correlation at an Unknown Date

- Tse (2000): a Lagrange multiplier test of H_0: constant correlation, against $\rho_{ij,t} = \rho_{ij} + \delta_{ij}\,\varepsilon_{i,t-1}\varepsilon_{j,t-1}$; computed from the fitted CCC model alone
- Picking a break date from a chart and then testing it with a fixed-date Fisher z test is oversized: the date is itself a function of the data
- Wied, Krämer & Dehling (2012): $Q_T = \max_{2 \le j \le T} \frac{j}{\sqrt{T}\,\hat{D}} \left|\hat{\rho}_j - \hat{\rho}_T\right|$, $\hat{\rho}_j$ the correlation of the first j observations; estimated break date $\hat{j} = \arg\max$
  - Compares the correlation of the first j days with the full-sample $\hat{\rho}_T$
  - $\hat{D}^2$: the delta-method long-run variance of $\hat{\rho}$, Bartlett kernel with bandwidth $\lfloor T^{1/4} \rfloor$ ($\lfloor \cdot \rfloor$: integer part)
  - Under H_0 (constant correlation, means and variances): $Q_T \to \sup_{0 \le s \le 1} |B(s)|$, B a Brownian bridge (a Brownian motion tied to 0 at $s = 0$ and $s = 1$), $s = j/T$
  - 5% critical value 1.358
  - Constant variances are required, so we apply it to GARCH-standardised residuals

Dating the SPY–TLT Correlation Break (1/2)



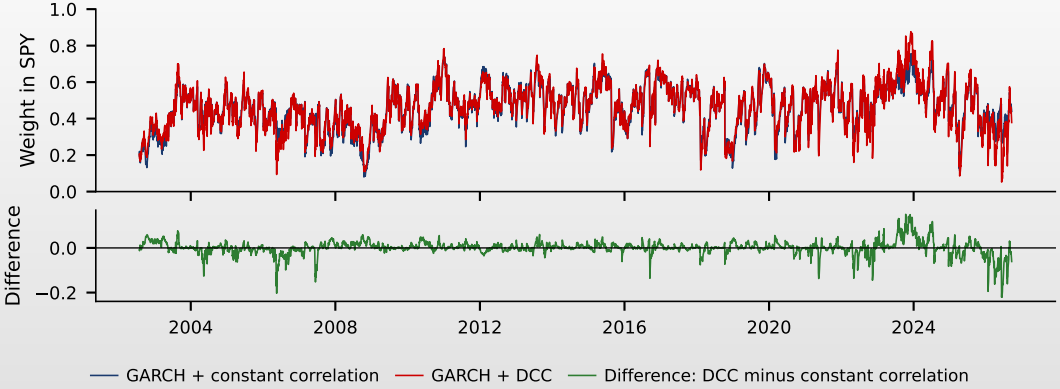
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Dating the SPY–TLT Correlation Break (2/2)

- Wied–Krämer–Dehling process on the GARCH(1,1)-standardised residuals, $T = 6\,073$, bandwidth 8
- Dashed: 5% critical value
- $Q_T = 4.25$ against 1.358: constant correlation rejected ($p < 0.001$)
- Estimated break 3 August 2020, not January 2022: residual correlation -0.36 before, 0.07 after
- Binary segmentation (the test repeated on each sub-period): before the main break, $Q = 3.01$ and a second break on 9 July 2007
- After the main break (August 2020 onwards): $Q = 1.56 > 1.358$, $p = 0.015$, so the correlation is not constant there either
- The limit law is proved for observed series: on estimated residuals the p-values are approximate
- One DCC target over 2002–2026 averages at least three regimes (breaks in July 2007 and August 2020)

MFM_ch6_corr_inference

Application: Minimum-Variance Weights with DCC (1/2)



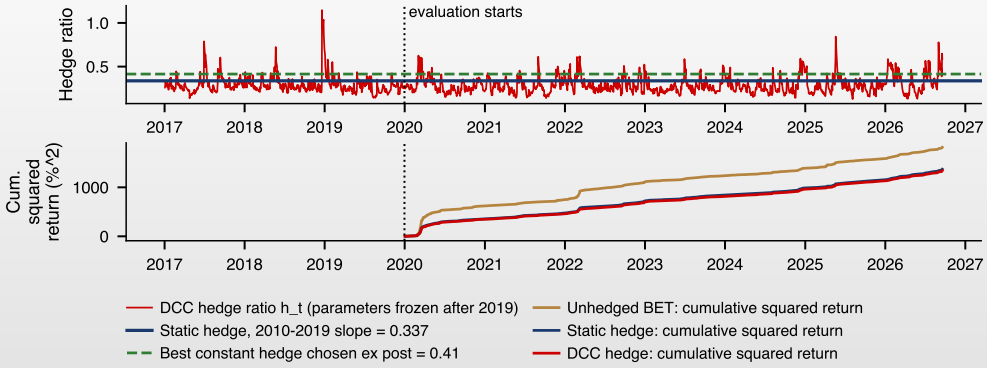
MFM_ch6_dcc_minvar

Application: Minimum-Variance Weights with DCC (2/2)

- Weight in SPY of the daily minimum-variance SPY–TLT portfolio, $w_t = (\sigma^2_{2t} - \sigma_{12,t})/(\sigma^2_{1t} + \sigma^2_{2t} - 2\sigma_{12,t})$, truncated to $[0, 1]$, $\sigma_{12,t} = \rho_t \sigma_{1t} \sigma_{2t}$ the conditional covariance
- Same GARCH volatilities, correlation from DCC or constant, equal to the full-sample 2002–2026 return correlation (-0.30)
- Annualised volatility: 60/40 fixed 10.99%, GARCH with constant correlation 9.10%, GARCH with DCC 9.06%
- Most of the gain comes from GARCH volatilities
- DCC adds little. Mean absolute weight gap DCC minus constant 0.019, largest 0.22 (8 June 2026)
- In-sample illustration: a fair test re-estimates on expanding windows and adds transaction costs (Chapter 4; out-of-sample evaluation: Chapter 8)

MFM_ch6_dcc_minvar

Dynamic Hedging Out of Sample: BET with Euro Stoxx 50 (1/2)



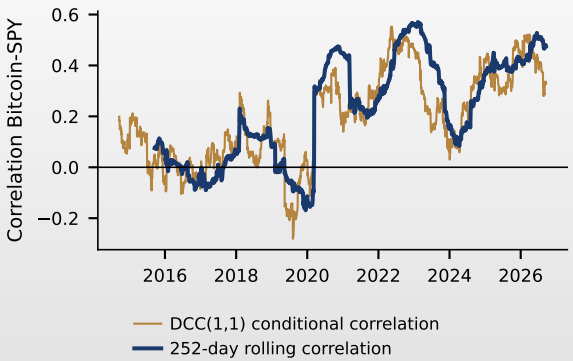
MFM_ch6_corr_inference

Dynamic Hedging Out of Sample: BET with Euro Stoxx 50 (2/2)

- The BET investor hedges with the Euro Stoxx 50: there are no liquid BET futures, and the two markets close almost synchronously
- Daily estimation from 5 January 2010 to the end of 2019 ($n = 2\,491$)
- Out-of-sample evaluation in 2020–2026 ($n = 1\,663$) with frozen parameters; the top panel starts in 2017
- Static hedge: the slope in the estimation window, $h = 0.337$
- DCC hedge: $h_t = \rho_t \sigma_{S,t} / \sigma_{F,t}$ (S: BET, F: Euro Stoxx 50, $\hat{a} = 0.009$, $\hat{b} = 0.964$), mean 0.29, range 0.13 to 0.84 in 2020–2026
- Variance removed in 2020–2026: 24.7% with the static hedge, 26.2% with the DCC hedge and 25.6% with the best constant hedge chosen ex post ($h = 0.41$)
- The Diebold & Mariano (1995) test (a t-test on the mean loss difference d_t = squared hedged return, static minus DCC, with a HAC SE): $t = 1.29$, one-sided $p = 0.098$, so the DCC gain is small and not significant at 5%

MFM_ch6_corr_inference

Bitcoin and US Equities: DCC Correlation



- Bitcoin and SPY on common trading days since 18 September 2014 ($T = 3\,018$)
- Weekend Bitcoin moves enter the Monday return
- Average DCC correlation 0.04 before 2020, 0.31 since January 2020
- Peak 0.55 on 19 May 2022
- Very persistent: $a = 0.015$, $b = 0.982$, $a + b = 0.9968$
- LR 105.6; bootstrap under CCC ($B = 199$): $p = 1/(B + 1) = 0.005$, the smallest attainable value: CCC rejected
- From diversifier to risk asset: the full-sample correlation (0.24) hides two regimes
- Liu & Tsyvinski (2021): until 2018 cryptos had small exposures to stock-market factors; after 2020 DCC shows the opposite

MFM_ch6_dcc_btc_spy

Contagion or Interdependence?



Lehman Brothers headquarters, New York, 15 September 2008, the day of the bankruptcy filing

Photo: Robert Scoble (2008); CC BY 2.0; Wikimedia Commons

- Common finding: correlations measured during crises are higher than in calm periods
- A high crisis correlation is not yet contagion, it must exceed the calm one
- Two very different explanations
  - **Contagion**: the transmission mechanism itself changes
  - **Volatility bias**: the mechanism is unchanged, but a more volatile source market mechanically raises the correlation
- Forbes & Rigobon (2002): after correcting the bias, most of the 1987, 1994 and 1997 "contagion" became interdependence
- Longin & Solnik (2001), Ang & Chen (2002): correlations rise in bear markets, not in bull markets, an asymmetry the Normal distribution cannot produce

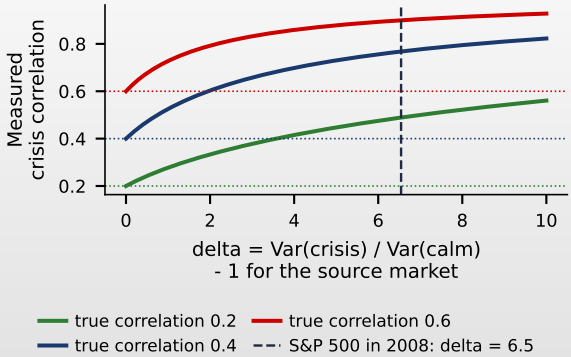
The Heteroskedasticity Bias (1/2): Correlation in a Calm Period

- Model: $y_t = \alpha + \beta x_t + \varepsilon_t$, $\mathrm{Cov}(x_t, \varepsilon_t) = 0$, β and σ_ε^2 constant
  - x_t: return of the source market
  - y_t: return of the other market
  - β: transmission coefficient
  - ε_t: own shock of y, variance σ_ε^2
  - σ_x^2: variance of x
- Two moments of the model
  - $\mathrm{Cov}(x_t, y_t) = \beta\sigma_x^2$ and $\mathrm{Var}(y_t) = \beta^2\sigma_x^2 + \sigma_\varepsilon^2$, because ε_t is uncorrelated with x_t
- Correlation: $\rho^2 = \dfrac{\beta^2\sigma_x^2}{\beta^2\sigma_x^2 + \sigma_\varepsilon^2} = \left(1 + \dfrac{\sigma_\varepsilon^2}{\beta^2\sigma_x^2}\right)^{-1}$
- The same in one number: the signal-to-noise ratio $k = \beta^2\sigma_x^2/\sigma_\varepsilon^2$
  - $\rho^2 = \dfrac{k}{1+k}$, equivalently $k = \dfrac{\rho^2}{1-\rho^2}$: the correlation depends on σ_x^2 only through k

The Heteroskedasticity Bias (2/2): From Calm to Crisis

- Crisis (subscript c): $\sigma^2_{x,c} = (1+\delta)\,\sigma^2_x$, $\delta > 0$ the relative rise of the source variance
- Nothing else changes
- Only σ^2_x grows, so $k_c = (1+\delta)\,k$ and
  - $\rho_c^2 = \dfrac{k_c}{1+k_c} = \dfrac{(1+\delta)\rho^2/(1-\rho^2)}{1+(1+\delta)\rho^2/(1-\rho^2)} = \dfrac{(1+\delta)\rho^2}{1+\delta\rho^2}$
- Then $\rho_c = \rho\sqrt{\dfrac{1+\delta}{1+\delta\rho^2}}$, so $|\rho_c| > |\rho|$ for $0 < |\rho| < 1$: the measured correlation rises in absolute value although β is the same
- Inverting: $\rho^* = \dfrac{\rho_c}{\sqrt{1+\delta(1-\rho_c^2)}}$ is the crisis correlation adjusted for the volatility increase
- Quick check: the source variance doubles ($\delta = 1$) and $\rho = 0.5$
  - $k = 0.25/0.75 = 1/3$, $k_c = 2/3$, $\rho_c^2 = (2/3)/(5/3) = 0.4$, so $\rho_c = 0.632$: a rise of 0.13 with no change in β
  - Adjusting: $\rho^* = 0.632/\sqrt{1+1\cdot(1-0.4)} = 0.632/1.265 = 0.5$, the calm value is recovered

How Much Can Volatility Alone Raise a Correlation?



- Measured crisis correlation ρ_c as a function of δ, for three true correlations that do not change
- Vertical line: the variance of 2-day S&P 500 returns in the 2008 crisis window was 7.5 times its calm-year level ($\delta = 6.54$)
- At this δ an unchanged true correlation of 0.4 is measured as 0.77, and 0.2 as 0.49

MFM_ch6_fr_bias

The Forbes–Rigobon Test and Its Inference

- Procedure
  - Choose the source market x, a calm window and a crisis window
  - Use 2-day returns (closing times differ)
  - Compute $\delta = \hat{\sigma}^2_{x,c}/\hat{\sigma}^2_{x,calm} - 1$ and ρ^*
  - Test $H_0 : \rho^* \leq \rho_{calm}$ with Fisher z statistics, the convention of Forbes & Rigobon (2002)
- Inference
  - With δ fixed, $\mathrm{Var}(\operatorname{artanh}\rho^*) \approx 1/[n_c(1+\delta(1-\rho_c^2))] < 1/n_c$ (n_c: crisis observations): the Fisher test is conservative
  - δ is estimated and 2-day returns overlap: resample the calm and crisis windows jointly (stationary bootstrap, mean block 10 days, $B = 1\,999$ draws) and test $D = \rho^* - \rho_{calm}$
  - Stationary bootstrap: it resamples blocks of consecutive days with random starts and geometric random lengths, which keeps the serial dependence inside each block

Assumptions to remember

- No omitted common shock: a global factor hitting both markets raises σ_ε as well as σ_x, and the adjustment over-corrects (Corsetti, Pericoli & Sbracia (2005); Appendix)
- No feedback from y to x (no endogeneity)
- The choice of windows matters: results can flip with a few weeks
- The correction goes too far if the mechanism really did change

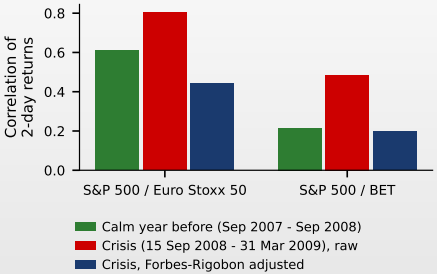
Worked Example: The 2008 Crisis Step by Step (1/2): Adjusting

- Source market: S&P 500
- Calm year: 14 September 2007 – 12 September 2008 ($n = 242$ 2-day returns)
- Crisis: 15 September 2008 – 31 March 2009 ($n = 130$)
- Variance ratio: $1 + \delta = 7.54$, so $\delta = 6.54$
- Euro Stoxx 50: $\rho_{calm} = 0.610$, $\rho_c = 0.804$
  - $\rho^* = 0.804/\sqrt{1 + 6.54\,(1 - 0.804^2)} = 0.442$
- BET: $\rho_{calm} = 0.216$, $\rho_c = 0.485$
  - $\rho^* = 0.485/\sqrt{1 + 6.54\,(1 - 0.485^2)} = 0.198$
- Both raw crisis correlations exceed the calm ones by far
- After the adjustment both fall below them

Worked Example: The 2008 Crisis Step by Step (2/2): Testing

- One-sided Fisher test of $H_0 : \rho_1 \leq \rho_{calm}$, with $\rho_1 = \rho_c$ (raw) or $\rho_1 = \rho^*$ (adjusted)
  - $z = \dfrac{\operatorname{artanh}\rho_1 - \operatorname{artanh}\rho_{calm}}{\sqrt{1/(n_c - 3) + 1/(n_{calm} - 3)}}, \quad p = 1 - \Phi(z)$
- 2-day returns overlap, so the sample sizes must be adjusted
  - We use effective sizes $n_{calm} = 121$ and $n_c = 65$, half of 242 and 130
  - For i.i.d. Normal daily pairs, the Bartlett approximation gives an effective size near $2n/3$, so halving is a cautious convention in that benchmark, not a general bound
- Euro Stoxx 50, raw: $z = \dfrac{\operatorname{artanh} 0.804 - \operatorname{artanh} 0.610}{\sqrt{1/62 + 1/118}} = 2.56$
  - One-sided p-value 0.005; with ρ^*, $z = -1.49$ and $p = 0.932$
- BET
  - $z = 1.97$ raw ($p = 0.024$), $z = -0.12$ adjusted ($p = 0.548$)
- Check: with all 130 and 242 overlapping returns the raw Euro Stoxx 50 statistic would be $z = 3.66$, $p \approx 0.0001$: treating overlapping returns as independent overstates the evidence

2008: Raw versus Adjusted Crisis Correlations



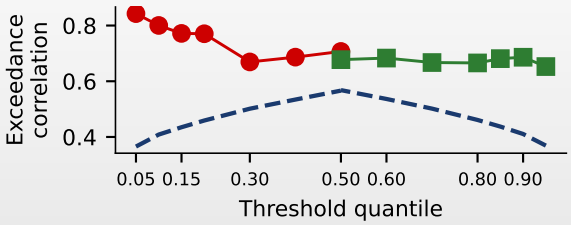
- Correlation of 2-day returns with the S&P 500: calm year, crisis (raw) and crisis after the Forbes–Rigobon adjustment
- Both raw rises are significant at 5%, neither survives the adjustment
- Stationary bootstrap: 90% interval for δ [3.9, 9.6]
- Euro Stoxx 50: 90% interval for ρ^* [0.38, 0.52], $p = 0.98$
- BET: 90% interval for ρ^* [0.16, 0.24], $p = 0.62$; bootstrap SE of ρ^* 0.025, against 0.047 from the Normal formula
- Verdict: no evidence of contagion once volatility is adjusted for, under the maintained assumptions
- Caveats: in 2008 a common global shock hit every market, so the adjustment may over-correct (Corsetti, Pericoli & Sbracia (2005)); the test also has low power (the 2020 application: Seminar B3)

MFM_ch6_forbes_rigobon

Exceedance Correlations

- The question of this section: how likely are joint extreme losses, and which dependence model reproduces them when correlation cannot?
- Longin & Solnik (2001): correlation conditional on both returns being beyond a threshold
- Lower tail at quantile q: $\rho^{-}(q) = \mathrm{Corr}\big(x, y \mid x \le F_x^{-1}(q),\, y \le F_y^{-1}(q)\big)$
- $F_x^{-1}(q)$: the q-quantile of x (e.g. $q = 0.10$: the worst 10% of weeks)
- Upper tail: the same with $x \ge F_x^{-1}(1-q)$, $y \ge F_y^{-1}(1-q)$
- Benchmark: for a bivariate Normal distribution, $\rho^{\pm}(q) \to 0$ as $q \to 0$
  - Conditioning on extremes lowers the correlation even when nothing changes: compare with the Normal curve, not with ρ

Exceedance Correlations: S&P 500 and Euro Stoxx 50



- Weekly returns 2000–2026 on the common trading days of the S&P 500 and the Euro Stoxx 50 ($n = 1\,393$, overall $\hat{\rho} = 0.78$)
- Dashed: the bivariate Normal distribution with the same correlation
- At $q = 0.10$: joint losses $\rho^{-} = 0.80$ (86 weeks), joint gains $\rho^{+} = 0.69$ (70 weeks)
- Normal benchmark 0.41
- Both tails lie far above the Normal benchmark, and crashes are shared more than rallies
- A single coefficient cannot describe this: we need a model of the whole dependence structure, a copula

MFM_ch6_exceedance

Why Correlation Is Not Enough

- Embrechts, McNeil & Straumann (2002): three fallacies about linear correlation
- Correlation measures only *linear* dependence and is not invariant to monotone transformations: $\text{Corr}(x, y) \neq \text{Corr}(e^x, e^y)$
- Margins and correlation do not determine the joint distribution
  - Two portfolios with identical margins and ρ can have very different Value-at-Risk (Chapter 7)
- Attainable correlations: for given margins, $\rho \in [\rho_{\min}, \rho_{\max}]$, reached by countermonotone and comonotone pairs
  - Lognormal $X = e^{\sigma_1 Z}$, $Y = e^{\pm\sigma_2 Z}$, Z standard Normal ($+$: comonotone, $-$: countermonotone): $\rho_{\max} = \dfrac{e^{\sigma_1\sigma_2} - 1}{\sqrt{(e^{\sigma_1^2} - 1)(e^{\sigma_2^2} - 1)}}$, $\rho_{\min}$ with $e^{-\sigma_1\sigma_2}$ in the numerator
  - $\sigma_1 = 1$, $\sigma_2 = 3$: $\rho_{\max} = 0.162$ and $\rho_{\min} = -0.008$: perfectly dependent, yet almost uncorrelated

Sklar's Theorem: Margins plus Copula

- A **copula** is a joint distribution function on $[0,1]^d$ with uniform margins
- Sklar (1959): for any joint distribution F with margins $F_1, \ldots, F_d$ there is a copula C with $F(x_1, \ldots, x_d) = C\big(F_1(x_1), \ldots, F_d(x_d)\big)$
  - d: number of variables
  - F_i: marginal distribution function of x_i
  - $u_i = F_i(x_i) \in [0,1]$: its probability level
  - C is unique if the margins are continuous
  - Conversely, any copula and any margins give a valid joint distribution
  - With densities: $f(x) = c\big(F_1(x_1), \ldots, F_d(x_d)\big) \prod_i f_i(x_i)$, c the copula density, f_i the marginal densities (the comonotone copula has none)
- Margins carry volatility, skewness and fat tails of each asset (Chapters 1 and 5)
- The copula carries all the dependence and is invariant to increasing transformations of each variable
- Three reference copulas and the Fréchet–Hoeffding bounds $W \le C \le M$ (Nelsen (2006))
  - Independence $\Pi(u,v) = uv$
  - Comonotonicity $M(u,v) = \min(u,v)$
  - Countermonotonicity $W(u,v) = \max(u+v-1, 0)$

Rank Correlations: Kendall and Spearman

- **Kendall's tau** (Kendall (1938)): concordant minus discordant pairs
  - $\tau = P\big[(X_1 - X_2)(Y_1 - Y_2) > 0\big] - P\big[(X_1 - X_2)(Y_1 - Y_2) < 0\big]$
  - (X_1, Y_1), (X_2, Y_2): two independent draws of the pair
  - Concordant: both coordinates move in the same direction
  - Copula form: $\tau = 4 \int\int C \, dC - 1 = 4\,\mathbb{E}[C(U, V)] - 1$
- **Spearman's rho** (Spearman (1904)): Pearson correlation of the ranks
  - Copula form: $\rho_S = 12 \int\int (C(u, v) - uv) \, du \, dv$
- Both lie in $[-1, 1]$ and depend only on the copula: 1 for M, 0 for Π, -1 for W
- θ: the copula parameter

Links to copula parameters

Copula	τ
Gaussian, t	$\frac{2}{\pi} \arcsin \rho$
Clayton	$\theta/(\theta + 2)$
Gumbel	$1 - 1/\theta$
Frank	$1 - \frac{4}{\theta}\left[1 - D_1(\theta)\right]$

- $D_1(\theta) = \frac{1}{\theta} \int_0^\theta \frac{t}{e^t - 1} dt$ (Debye function)
- Gaussian: $\rho_S = \frac{6}{\pi} \arcsin(\rho/2)$
- Example, Gaussian copula with $\rho = 0.5$: $\tau = \frac{2}{\pi} \cdot \frac{\pi}{6} = 1/3$ and $\rho_S = \frac{6}{\pi} \arcsin 0.25 = 0.483$

Five Copula Families

Family	$C(u, v)$	Parameter
Gaussian	$\Phi_{\mathbf{2},\rho}\left(\Phi^{-\mathbf{1}}(u), \Phi^{-\mathbf{1}}(v)\right)$	$-1 < \rho < 1$
Student t (Demarta & McNeil (2005))	$t_{\mathbf{2},\rho,\nu}\left(t_\nu^{-\mathbf{1}}(u), t_\nu^{-\mathbf{1}}(v)\right)$	$-1 < \rho < 1,\ \nu > 0$
Clayton (Clayton (1978))	$\left(u^{-\theta} + v^{-\theta} - 1\right)^{-\mathbf{1}/\theta}$	$\theta > 0$
Gumbel (Gumbel (1960))	$\exp\left(-\left[(-\ln u)^\theta + (-\ln v)^\theta\right]^{\mathbf{1}/\theta}\right)$	$\theta \geq 1$
Frank (Frank (1979))	$-\frac{\mathbf{1}}{\theta}\ln\left(1 + \frac{(e^{-\theta u}-\mathbf{1})(e^{-\theta v}-\mathbf{1})}{e^{-\theta}-\mathbf{1}}\right)$	$\theta \neq 0$

- $\Phi, \Phi^{-\mathbf{1}}$, $\Phi_{\mathbf{2},\rho}$: the Normal distribution function, quantile and bivariate distribution function (correlation ρ); t_ν, $t_{\mathbf{2},\rho,\nu}$: the Student t analogues with ν degrees of freedom; θ: dependence parameter
- Why these five: they cover the four tail patterns: none (Gaussian, Frank), both symmetric (t), lower tail only (Clayton), upper tail only (Gumbel)
- Elliptical (Gaussian, t): extend to d dimensions with a correlation matrix, the standard choice in risk systems
- $\nu \to \infty$ turns the t copula into the Gaussian one
- Archimedean (Clayton, Gumbel, Frank): $C(u, v) = \varphi^{-\mathbf{1}}\left(\varphi(u) + \varphi(v)\right)$, φ: generator (decreasing, $\varphi(1) = 0$)
  - One parameter, closed form, easy simulation, but exchangeable (symmetric in the variables, $C(u, v) = C(v, u)$) and poor in high dimension
- Rotation by 180° (the survival copula of $(1 - U, 1 - V)$) moves the tail dependence to the opposite corner (densities and generators in the ▸ Appendix)

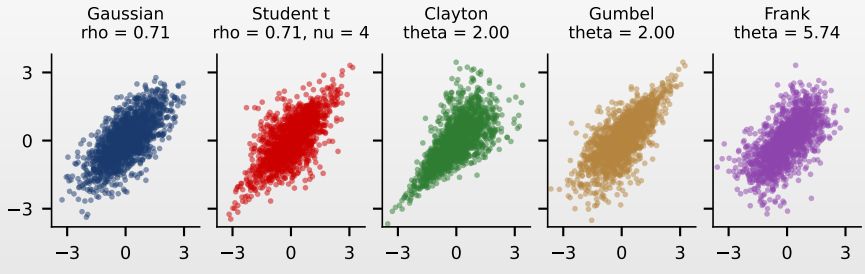
Tail Dependence

- Lower tail:
  $\lambda_L = \lim_{q \downarrow \mathbf{0}} P(V \le q \mid U \le q) = \lim_{q \downarrow \mathbf{0}} C(q,q)/q$, with
  $U = F_{\mathbf{1}}(X_{\mathbf{1}})$, $V = F_{\mathbf{2}}(X_{\mathbf{2}})$ and q the tail probability
- Upper tail: $\lambda_U = \lim_{q \downarrow \mathbf{0}} P(V > 1 - q \mid U > 1 - q)$
- $\lambda > 0$: the probability of a joint extreme stays positive however extreme the threshold; $\lambda = 0$: asymptotic independence
- Tail dependence is a copula property, invisible in ρ
- At a finite q, $C(q,q)/q$ is a probability, not yet the limit
- $t_{\nu+\mathbf{1}}$ in the table: Student t distribution function with $\nu + 1$ degrees of freedom

Closed forms

	λ_L	λ_U
Gaussian	0	0
Student t	$2\, t_{\nu+\mathbf{1}}\Big(-\sqrt{\frac{(\nu+\mathbf{1})(\mathbf{1}-\rho)}{\mathbf{1}+\rho}}\Big)$	same
Clayton	$2^{-\mathbf{1}/\theta}$	0
Gumbel	0	$2 - 2^{\mathbf{1}/\theta}$
Frank	0	0

Five Copulas with the Same Kendall's Tau



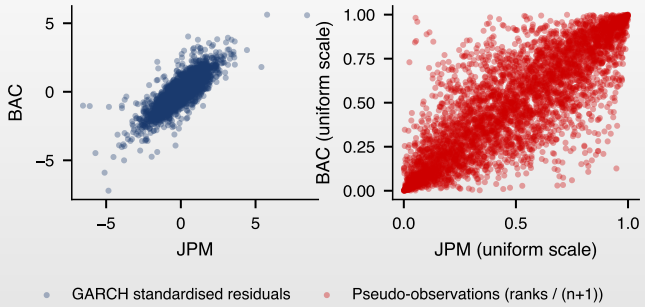
- 1 500 simulated pairs from each copula with $\tau = 0.5$ on Normal-score margins $(\Phi^{-1}(u), \Phi^{-1}(v))$
- Gaussian and t $\rho = 0.707$ (t with $\nu = 4$), Clayton and Gumbel $\theta = 2.0$, Frank $\theta = 5.74$
- Corners: Gaussian and Frank thin out, t stays dense in both, Clayton is sharp lower-left (joint crashes), Gumbel sharp upper-right
- Equal Kendall dependence does not determine joint-tail risk

MFM_ch6_copula_zoo

Estimating Copulas

- **Full maximum likelihood**: margins and copula estimated jointly
  - Efficient, but computationally demanding and sensitive to misspecification
- **IFM** (inference functions for margins, Joe (2005)): two steps
  - Fit parametric margins (e.g. GARCH with Student t)
  - Set $\hat{u}_{it} = \hat{F}_i(x_{it})$ ($\hat{F}_i$: fitted marginal distribution function)
  - Maximise $\sum_t \ln c(\hat{\boldsymbol{u}}_{\mathbf{1}t}, \hat{\boldsymbol{u}}_{\mathbf{2}t}; \theta)$, c the copula density
- **CML** (canonical maximum likelihood, Genest, Ghoudi & Rivest (1995)): no parametric margins
  - Pseudo-observations $\hat{u}_{it} = \text{rank}(x_{it})/(T+1)$
  - Then maximise the copula likelihood
  - Robust to misspecified margins
  - The $T+1$ keeps $\hat{u}$ away from 0 and 1
- For returns: filter each series with GARCH first, because the copula assumes i.i.d. observations and volatility clustering would mix dependence with changing volatility; then use the ranks of the standardised residuals (CML, the method used from here on)
- Model choice: the families are not nested, so the LR test does not apply; we use AIC $= 2k - 2\ell$ (Akaike information criterion), BIC $= k \ln T - 2\ell$ (Bayesian information criterion)
- k: number of parameters, ℓ: maximised log-likelihood

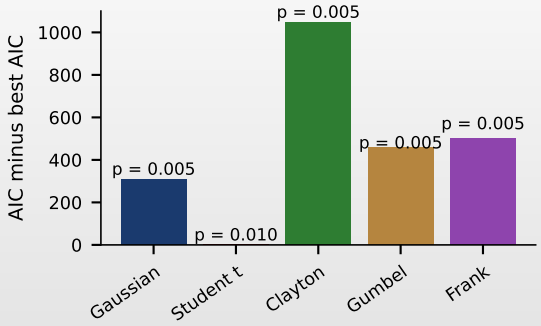
From Residuals to Pseudo-Observations: JPM and BAC



- JPMorgan Chase (JPM) and Bank of America (BAC), daily 2010–2026 ($T = 4\,202$), GARCH(1,1)-filtered
- Kendall's $\hat{\tau} = 0.641$, Spearman's $\hat{\rho}_S = 0.822$
- Both corners of the uniform scatter are dense: joint crashes and joint rallies

MFM_ch6_copula_banks

Copula Estimates and Goodness of Fit: JPM and BAC



Numbers above bars: goodness-of-fit p-value
(Rosenblatt Cramer-von Mises, parametric bootstrap)

	$\hat{\theta}$ or $\hat{\rho}$	$\hat{\nu}$	λ_L	λ_U
Gaussian	0.833	–	0.000	0.000
Student t	0.842	4.24	0.531	0.531
Clayton	2.384	–	0.748	0.000
Gumbel	2.587	–	0.000	0.693
Frank	9.152	–	0.000	0.000

- The bars show the AIC distance, the numbers above them the bootstrap goodness-of-fit p-values (full table in the Appendix)
- Student t has the lowest AIC, by a wide margin
- The Rosenblatt test (below) rejects every family: each p-value is the smallest attainable, $1/(B+1)$
- Statistics: t 0.082, Frank 0.365, Gaussian 0.968, Clayton 1.24, Gumbel 1.42; Frank is second by fit but fourth by AIC

MFM_ch6_copula_banks

Inference for Copula Parameters (1/2): Inverting Kendall's Tau

- **Inversion of** $\hat{\tau}$: a U-statistic with a Hoeffding-projection standard error
- $\hat{\theta} = g(\hat{\tau})$ by the delta method (Genest & Rivest (1993))
  - g: the inverse of the link $\tau(\theta)$ of the previous table
  - U-statistic: an average over all pairs of observations
  - Elliptical copulas: $\tau = \frac{2}{\pi} \arcsin \rho$ for *every* ν, so τ identifies ρ, never ν
  - JPM–BAC: $\hat{\tau} = 0.641$ (SE 0.0065), $\hat{\rho}_\tau = \sin(\pi\hat{\tau}/2) = 0.845$ (SE 0.0055)
  - CML $\hat{\rho} = 0.842$ (SE 0.0053)

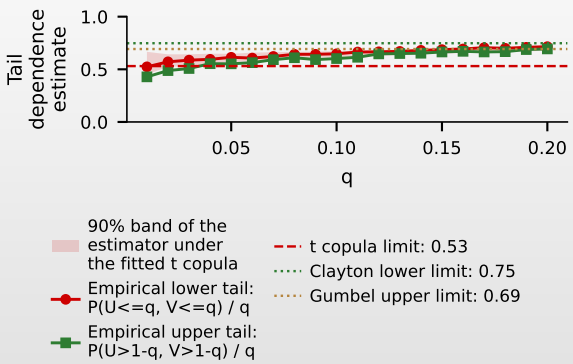
Inference for Copula Parameters (2/2): Ranks, Fit and GARCH Filtering

- **CML: the ranks are estimated**, so the inverse Hessian is too small
- The Genest, Ghoudi & Rivest (1995) sandwich adds the rank terms
  - Increase of the SE: +60% for the Gaussian ρ (S&P 500–Euro Stoxx 50 weekly, 0.0086 to 0.0138), +17% for the t-copula ρ (0.0046 to 0.0053) and +1% for ν (JPM–BAC)
- Not all of the +60% is the price of ranking
  - If the Gaussian copula were correct, estimating the margins by ranks would multiply the SE by $\sqrt{1+\rho^2}$ (Appendix)
  - At the weekly correlation 0.78 the factor would be 1.27; Monte Carlo at $\rho = 0.8$ gives 1.28
  - The observed factor, 1.60, points to a misspecified Gaussian copula
  - The sandwich stays valid in this case too (fit: Seminar B2)
- **After GARCH filtering** (Chen & Fan (2006)): CML on the ranks of estimated residuals has the same limit as on the true innovations
  - Conditions: i.i.d. innovations with copula C_θ and a correctly specified GARCH
  - Then the sandwich stays valid and, unlike DCC, no step-1 correction is needed (Monte Carlo in the Appendix)

Goodness of Fit: The Rosenblatt Test

- If $(U, V) \sim C_\theta$, then $E_1 = U$ and $E_2 = h(V \mid U) = \partial C_\theta(U, V)/\partial u$ (the conditional distribution function of V given U) are independent uniforms
- Statistic (Genest, Rémillard & Beaudoin (2009)): $S_n = n \int_{[0,1]^2} \left(D_n(e) - e_1 e_2\right)^2 de$
  - $e = (e_1, e_2) \in [0, 1]^2$
  - $D_n(e)$: share of the n pairs with $E_1 \leq e_1$ and $E_2 \leq e_2$
  - $e_1 e_2$: the same share under independence
  - $S_n \geq 0$, large values mean a poor fit
- p-value by parametric bootstrap, because θ is estimated
  - Simulate n pairs from $C_{\hat{\theta}}$, re-rank them into pseudo-observations, re-estimate, recompute $S_n^{(b)}$
  - $p = (1 + \#\{S_n^{(b)} \geq S_n\})/(B + 1)$
  - For JPM–BAC $B = 200$ (100 for Student t)
  - The smallest attainable p-value is therefore $1/201 \approx 0.005$ ($1/101 \approx 0.010$ for t)
- On GARCH residuals, Rémillard (2017) shows that rank-based tests keep their limit and the parametric bootstrap applies
- With $T = 4\,202$ the test has high power against small departures: a rejection does not say which feature is missing

Empirical Tail Dependence: JPM and BAC



- $\hat{\lambda}_L(q) = \hat{C}(q,q)/q$ and $\hat{\lambda}_U(q)$ for q from 1% to 20%
- Band: 90% range of the same estimator on data simulated from the fitted t copula
- $q = 5\%$: $\hat{\lambda}_L = 0.61$, $\hat{\lambda}_U = 0.55$
- $q = 1\%$: 0.52 and 0.43
- When JPM has one of its 5% worst days, BAC is in its own worst 5% about 61% of the time
- Lower tail inside the t-copula band (0.57–0.65 at 5%): the t copula describes the crashes well
- The upper tail (0.55 at 5%) lies below the t-copula band: joint rallies are weaker than the model predicts
- At $q = 1\%$ only 42 days remain in the tail, so the estimate is imprecise

MFM_ch6_copula_banks

Deeper study

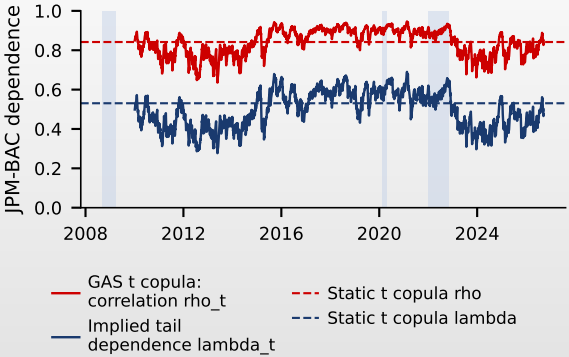
Is the Tail Asymmetry Real?

- Schmidt & Stadtmüller (2006): $\hat{\lambda}_L = \frac{n}{k}\,\hat{C}_n\left(\frac{k}{n}, \frac{k}{n}\right)$, the chart estimator with $q = k/n$
  - n: sample size
  - k: number of observations in the tail
  - $\hat{C}_n$: empirical copula
  - A larger k lowers the variance but raises the bias (conditions in the Appendix)
- $H_0 : \lambda_L(q) = \lambda_U(q)$ at a fixed q (threshold probabilities, not the limits)
- Bootstrap of residual pairs, re-ranked, $B = 999$

q	k	$\hat{\lambda}_L - \hat{\lambda}_U$	95% interval, i.i.d.	95% interval, blocks
5%	210	0.062	[−0.014, 0.148]	[−0.014, 0.138]
1%	42	0.095	[−0.095, 0.286]	[−0.119, 0.286]

- The block version (stationary bootstrap, mean block 1.6 and 1.0 days, Politis & White (2004) rule) allows serial dependence in the tail indicators
- All intervals contain zero: the asymmetry is not significant, so the symmetric t copula is not contradicted in the tails

A Dynamic t Copula for JPM and BAC (1/2)



MFM_ch6_copula_inference

A Dynamic t Copula for JPM and BAC (2/2)

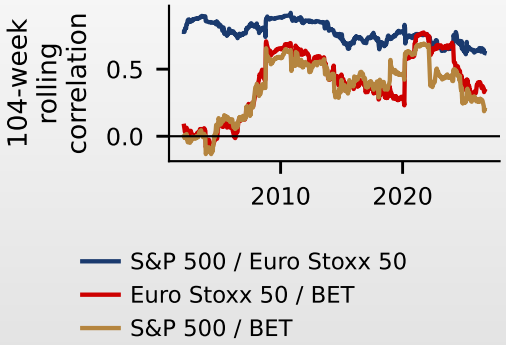
- GAS (generalised autoregressive score) t copula, Creal, Koopman & Lucas (2013): $\rho_t = \tanh f_t$, $f_{t+\mathbf{1}} = \omega + A\, s_t + B f_t$, same pseudo-observations
- f_t: the transformed correlation, on the whole real line; ω: constant; $s_t = \partial \ln c(\hat{u}_t, \hat{v}_t; \rho_t, \nu)/\partial f_t$: the score (unit scaling); A: reaction to the score; B: persistence
- Estimates: $A = 0.025$, $B = 0.9926$, $\nu = 5.35$ (static copula $\nu = 4.24$)
- ρ_t from 0.64 (23 May 2013) to 0.95 (18 February 2021); λ_t from 0.28 to 0.69, against a static 0.53; shaded: autumn 2008 – March 2009, February–April 2020, January–October 2022
- Once ρ moves, $\hat{\nu}$ rises: part of the static tail dependence was time variation in dependence
- LR against the static t copula: 245.5; under $A = 0$, B is not identified, so the $\chi^{\mathbf{2}}$ limit does not apply and the p-value comes from a bootstrap (Seminar B6)
- Time variation is one explanation consistent with the rejection of every static family

MFM_ch6_copula_inference

A Cautionary Tale: The Gaussian Copula and 2008

- Li (2000) modelled joint default times with a Gaussian copula
  - One correlation parameter, calibrated to market spreads
  - It priced whole credit portfolios and became the standard for tranches of collateralised debt obligations (CDO)
- Zero asymptotic tail dependence: for the most extreme joint defaults it gives lower probabilities than tail-dependent models calibrated to the same data
- A static correlation calibrated in calm years: exactly the Forbes–Rigobon and DCC lessons of this chapter

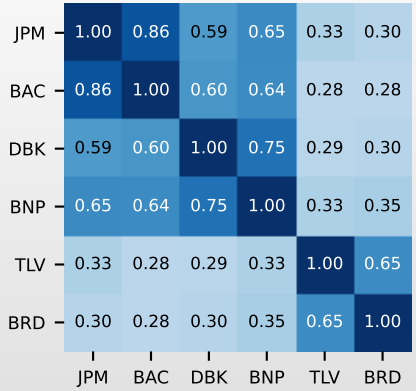
US, Euro-Area and Romanian Equities, 2000–2026



- The question of this section: is the Romanian market increasingly integrated with the large ones, and can one still diversify across banking systems in crises?
- Weekly returns, 104-week window, common trading days, Friday closes
- Euro Stoxx 50–BET: 0.33 in 2000–2009, 0.45 in 2010–2019, 0.61 since 2020
- S&P 500–Euro Stoxx 50: 0.84, 0.78, 0.70
- Rolling Euro Stoxx 50–BET: 0.31 at end-2007, 0.65 at end-2009, 0.29 at end-2019, 0.76 at end-2021, 0.35 in September 2026
- Jumps in crises and partial reversal afterwards: integration or the Forbes–Rigobon volatility bias? (Seminar C2)

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Banks: Correlations of Weekly Returns, 2010–2026 (1/2)



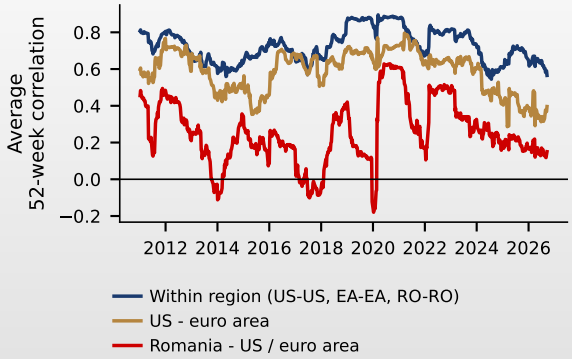
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Banks: Correlations of Weekly Returns, 2010–2026 (2/2)

- US: JPM and BAC
- Euro area: DBK (Deutsche Bank) and BNP (BNP Paribas)
- Romania: TLV (Banca Transilvania) and BRD (BRD-Groupe Société Générale)
- 871 weeks from 15 January 2010
- The choice: two of the largest listed banks of each system, with continuous data since 2010
- Three regional blocks: US 0.86, euro area 0.75, Romania 0.65
- Cross-region links are weakest for Romania: BNP–TLV 0.33, JPM–BRD 0.30
- Bank returns share country-level factors: sovereign risk, domestic rates, regulation

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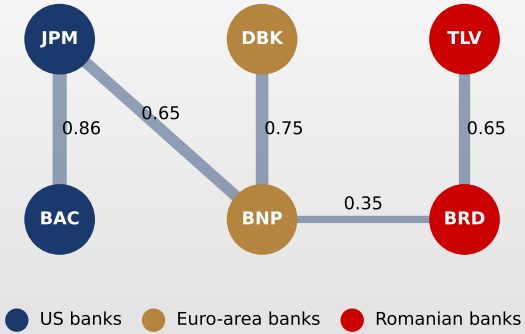
Bank Correlations over Time



- Average pairwise 52-week correlation of weekly returns: within regions (legend: US, EA: euro area, RO: Romania), between the US and the euro area, between Romania and the others
- Full period: within regions 0.73, US–euro area 0.59, Romania–others 0.25
- March–June 2020: 0.87, 0.69 and 0.56: the Romanian link more than doubled in a global shock
- Latest 0.57, 0.40, 0.15
- Diversification across banking systems works in normal times and weakens exactly in systemic events

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A Correlation Network of Banks



- Minimum spanning tree (Mantegna (1999)) on $d_{ij} = \sqrt{2(1-\rho_{ij})}$: the 5 strongest links that connect all 6 banks without cycles
- Edges: JPM–BAC (0.86); JPM–BNP (0.65); DBK–BNP (0.75); BNP–BRD (0.35); TLV–BRD (0.65)
- Romania is attached through BNP–BRD
- The distance $\sqrt{(1-\rho)/2}$ of Chapter 4 gives the same tree (both distances increase with $1-\rho$)
- The tree is undirected and contemporaneous: it does not identify transmission paths, only hypotheses for the network measures of Chapter 18 (literature in the Appendix)

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Case Study (1/2): Which Events Shake All Markets at Once?

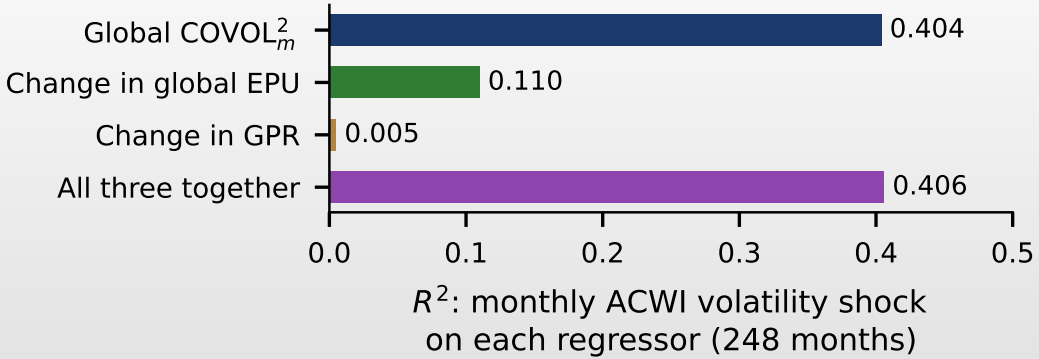
- Questions of Engle & Campos-Martins (2023)
  - which events move the volatilities of almost all countries and asset classes on the same day?
  - how large is this common shock?
- The daily size of the common shock is global COVOL (common volatility)
- The distinct idea: a correct DCC gives residuals e_t with $E_{t-\mathbf{1}}(e_t e_t^\top) = I$, but uncorrelated is not independent: the squares can comove
  - $\Psi_t = E_{t-\mathbf{1}}\left[(e_t^{\mathbf{2}} - \iota)(e_t^{\mathbf{2}} - \iota)^\top\right]$
  - e_t: vector of the N DCC residuals of day t
  - $e_t^{\mathbf{2}}$: its element-by-element square
  - ι: vector of ones
  - $E_{t-\mathbf{1}}$: expectation given the information at $t-1$
  - Off-diagonal elements $\Psi_{ij,t} \neq 0$: the squared shocks of assets i and j are large on the same days
  - The cross-sectional analogue of ARCH (autoregressive conditional heteroskedasticity, Chapter 5): correlated squares across assets on one day, not over time
  - Herskovic, Kelly, Lustig & Van Nieuwerburgh (2016) document such a factor in the idiosyncratic volatility of US stocks

Deeper study

Case Study (2/2): One Common Factor Scales Every Variance

- Model: one common factor x_t scales the variance of every residual
  - $e_{it} = \sqrt{s_i x_t + 1 - s_i}\,\varepsilon_{it}$, ε_{it} i.i.d. $\mathcal{N}(0,1)$, $x_t > 0$, $E_{t-\mathbf{1}}(x_t) = 1$
  - x_t: the squared global COVOL of day t ($x_t > 1$: a common shock day)
  - $s_i \in [0,1]$: loading of asset i, the weight of x_t in its conditional variance $g_{it} = s_i x_t + 1 - s_i = 1 + s_i(x_t - 1)$
- Estimation: only the products $s_i(x_t - 1)$ are identified, so the scale must be fixed by a normalisation
  - Equal loadings $s_i = 1$: $\hat{x}_t = \frac{1}{N}\sum_i e_{it}^2$, the cross-sectional mean of the squared residuals, on the variance scale of the residuals
  - Unequal loadings: alternating maximum likelihood over x_t (one cross-section per day) and s_i (one time series per asset), with $s^\top s = 1$ and mean $x_t = 1$
  - Under $s^\top s = 1$ the loadings are relative sizes, not variance shares (equal loadings would be $s_i = 1/\sqrt{N}$)
- Test ξ: the standardised mean over days and pairs of the products $(e_{it}^2 - 1)(e_{jt}^2 - 1)$; under H_0: $\Psi_{ij} = 0$, $i \neq j$, it is standard Normal
- Data: 47 country ETFs (exchange-traded funds) listed on the NYSE (New York Stock Exchange), 2000–2021, and 23 US asset-class series (details in the Appendix)

Realised Shocks against News Indices (1/2)



MFM_ch6_covol

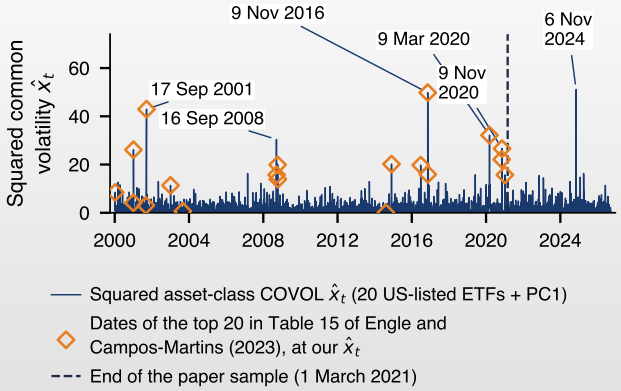
Deeper study

Realised Shocks against News Indices (2/2)

- Monthly, 248 months (Table 8 of Engle & Campos-Martins (2023)): ACWI (MSCI All Country World Index) volatility shock regressed on each index
  - On COVOL_m^2, the monthly mean of $\hat{x}_t$ minus 1: $R^2 = 0.404$ (R^2: share of variance explained)
  - On the change in EPU (economic policy uncertainty, Baker, Bloom & Davis (2016)): 0.110
  - On GPR (geopolitical risk, Caldara & Iacoviello (2022)): 0.005
  - Both news indices added: only 0.406
- Top days in the paper: 9 March 2020 ($\hat{x}_t = 51.4$) and 24 June 2016 (Brexit)
- 28 April 2020 is also a top day, but with a positive mean return (+2.33%)
  - COVOL measures the size of the common shock, not its sign
  - News indices measure what people fear, COVOL the shocks that hit prices (event table in the Appendix)

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Asset-Class COVOL on the Course ETFs, 2000–2026 (1/2)



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Deeper study

Asset-Class COVOL on the Course ETFs, 2000–2026 (2/2)

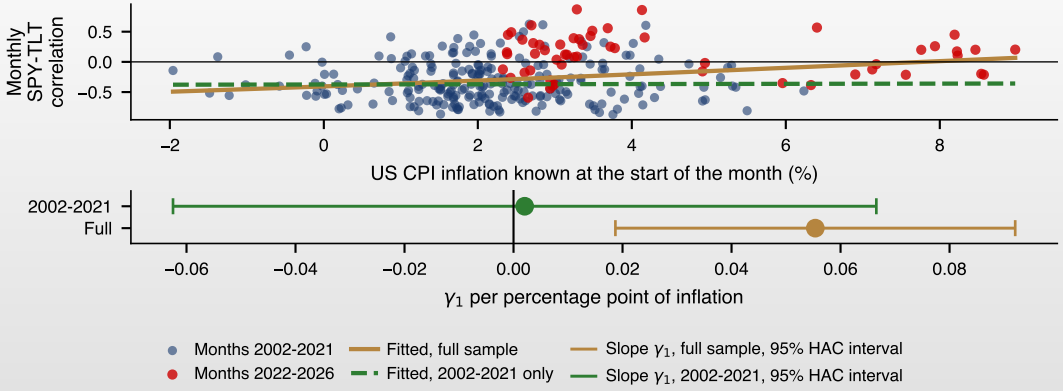
- 20 US-listed ETFs and their PC1 (first principal component), 4 January 2000 – 18 September 2026, 6 717 days
- AR(1) mean (first-order autoregressive) with PC1, GARCH(1,1), then the alternating estimation
- Diamonds: the 20 dates of Table 15 of Engle & Campos-Martins (2023), placed at our replication's value on that day (days with the market closed: at the next trading day)
- Top days up to 1 March 2021: 9 November 2016 ($\hat{x}_t = 49.8$), 17 September 2001 (42.9), 9 March 2020 (32.2): the 2016 US election, the reopening after 9/11 and the COVID-19 crash
- 13 of the 20 largest days of our replication before March 2021 are among the 20 dates of Table 15
- With $s^\top s = 1$ and mean $\hat{x}_t = 1$ the scale depends on the panel
- Replication values are not comparable with the paper's: compare rankings, not levels
- Largest in the full sample: 6 November 2024 (51.1, mean ETF return 0.95%), the day after the 2024 US election: a large common shock on an up day

MFM_ch6_covol

An Open Question: Will the Stock–Bond Correlation Stay Positive?

- The question: is the positive stock–bond correlation since 2022 a durable regime governed by inflation, and can it be forecast?
- Campbell, Pflueger & Viceira (2020): the sign of the comovement follows the inflation–output relation and the monetary-policy regime; Molenaar, Sénéchal, Swinkels & Wang (2024) extend the evidence across countries and a long history
- The facts (earlier sections): the SPY–TLT correlation is positive since 2022, and the break is dated August 2020
- Hypothesis: $z_m = \operatorname{artanh} \rho_m = \gamma_0 + \gamma_1 \pi_{m-2} + u_m$, with ρ_m the monthly realised correlation, π the annual US CPI (consumer price index) inflation and u_m the error
- H_0: $\gamma_1 = 0$ against H_1: $\gamma_1 > 0$, with a HAC t-statistic
- Falsified if $\gamma_1 \approx 0$ in periods where inflation varied but the regime did not
- The loop: an AI assistant drafts the literature map, competing hypotheses, the code, a robustness plan and a referee critique; the researcher resolves every DOI, replicates the chapter's numbers first and fixes the plan before the first run (prompts in the Appendix)

Mini-Case: Does Inflation Drive the Stock–Bond Correlation? (1/2)



MFM_ch6_ai_inflation

Mini-Case: Does Inflation Drive the Stock–Bond Correlation? (2/2)

- Monthly SPY–TLT correlation from daily returns against the inflation known at the start of month m: CPI for month $m-1$ is released only in mid-month m, so the latest known value is $\pi_{m-\mathbf{2}}$
- Current CPI vintage, so a retrospective test, not a real-time one
- Full sample, 287 months: $\hat{\gamma}_{\mathbf{1}} = 0.055$ per percentage point, HAC SE 0.019, $t = 2.96$, $R^{\mathbf{2}} = 0.047$
- Mean ρ -0.12 when inflation exceeds 3% (89 months), -0.30 otherwise
- 2002–2021 only (232 months, 53 with inflation above 3%): $\hat{\gamma}_{\mathbf{1}} = 0.002$, $t = 0.06$
- The falsification criterion is met on 2002–2021: inflation varied and $\hat{\gamma}_{\mathbf{1}} \approx 0$
- The full-sample slope rests on the post-2022 episode, and a single regime change cannot separate the inflation level from the policy regime (Campbell, Pflueger & Viceira (2020))

MFM_ch6_ai_inflation

What the Researcher Checks, and a Project Seed

Checks before any claim

- Look-ahead bias: release calendar and data vintages (the lag handles publication delay, not revisions)
- Calendars: align prices on common trading days before computing returns
- Persistent regressor, few regimes: subsamples, long HAC lags, a placebo with simulated persistent series
- Volatility bias: control for realised volatility
- Multiple testing: fix measures and windows in advance
- Invented references: an unresolvable DOI is dropped and logged in `AI_ERRORS.md`

Your turn: a team-project seed

- Does a DCC whose target depends on inflation and the policy rate forecast the SPY–TLT correlation better out of sample than the plain DCC?
- Replicate first: $a + b = 0.986$, $\bar{Q}_{\mathbf{12}} = -0.25$ and the slope $\hat{\gamma}_{\mathbf{1}} = 0.055$
- Deliverables: a one-page pre-registration, code on the course market data plus CPI, a Diebold–Mariano comparison and a referee-style self-critique
- The AI-use statement, in `AI_USE.md`

Decision Checklist: Question, Estimator, Uncertainty, Failure

Question	Estimator	Uncertainty	Fails when
Did a correlation change?	Pearson on aligned prices	HAC delta method, block bootstrap	Fisher under clustering; daily data across time zones
Covariance for risk today?	EWMA, DCC	bootstrap of both steps	the target $\bar{Q}$ moves with the regime
When did it break?	Wied–Krämer–Dehling on GARCH residuals	Brownian-bridge limit	estimated GARCH residuals (approximate p-value); unmodelled variances
Contagion?	Forbes–Rigobon ρ^*	stationary bootstrap, δ estimated	a common shock or feedback
Joint extremes?	copula on GARCH pseudo-observations	rank sandwich, parametric bootstrap	Gaussian copula ($\lambda = 0$), static copula when dependence moves

- Next: Chapter 7 turns DCC covariances and simulated copulas into portfolio VaR and ES

Check Your Understanding

- Why is the Fisher interval for the SPY–TLT correlation too narrow on daily data, and which two features does the HAC interval take into account?
- Why is the same-day correlation of SPY and Euro Stoxx 50 biased towards zero, and how do 2-day or weekly returns reduce the bias?
- Which source of uncertainty do step-2 DCC standard errors ignore, and how does the bootstrap of both steps restore it?
- A crisis correlation is higher than the calm one: what must you compute before calling the rise contagion, and which assumption can make the adjustment excessive?
- Five copulas share Kendall's $\tau = 0.5$: why can their probabilities of joint crashes still differ?

Further Study — Courses, Code and Data

Advanced courses on quantinar.com

- Measuring Statistical Risk
- Multivariate Statistical Analysis
- Minimum Spanning Tree
- FRM: Financial Risk Meter

Code, data and textbooks

- All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_06
- Textbooks: Franke, Härdle & Hafner (2019); Härdle & Simar (2019); Nelsen (2006)

Back

Inference on a Correlation, the Delta Method

- $\hat{\rho} = g(\hat{m})$, with $\hat{m}$ the sample mean of $m_t = (x_t,\ y_t,\ x_t^2,\ y_t^2,\ x_t y_t)$ and g the function that turns these five moments into the correlation; ∇g: its gradient
- Delta method: $\sqrt{T}(\hat{\rho} - \rho) \to N\big(0,\ \nabla g^\top \Omega\, \nabla g\big)$, Ω the long-run covariance of m_t, estimated by HAC with a Bartlett kernel and L lags (Newey & West (1987))
- The equivalent influence-function form, $\hat{\rho} - \rho \approx \frac{1}{T}\sum_t \psi_t$, is the one of the main text
- Conditions: stationarity, finite moments of order $4 + \delta$ of (x_t, y_t) and weak dependence (mixing); with infinite fourth moments the Normal limit fails
- The HAC interval, built on the Fisher scale: $\tanh\big(\operatorname{artanh}\hat{\rho} \pm 1.96\,\widehat{\mathrm{SE}}_{\mathrm{HAC}}/(1-\hat{\rho}^2)\big)$, with $\widehat{\mathrm{SE}}_{\mathrm{HAC}} = \sqrt{\nabla\hat{g}^\top \hat{\Omega}\, \nabla\hat{g}/T}$
- The classic Fisher interval replaces $\widehat{\mathrm{SE}}_{\mathrm{HAC}}/(1-\hat{\rho}^2)$ by $1/\sqrt{T-3}$
- The last 252-day window (the 5-day series) ends on 17 September 2026: $\hat{\rho} = 0.245$, Fisher interval [0.13, 0.36], HAC interval [0.07, 0.41]; largest HAC-to-Fisher width ratio: 2.02

Back

The VEC and BEKK Models

- **VEC** (Bollerslev, Engle & Wooldridge (1988)): in the equation of the main text, c is a constant vector
- VEC: $N(N+1)/2 \cdot [1 + N(N+1)]$ parameters
- **BEKK** (Engle & Kroner (1995)): positive definite by construction if $CC^\top$ is positive definite
- BEKK: $N(N+1)/2 + 2N^2$ parameters, hard to interpret, and the likelihood is flat in high dimension
- DCC: $3N + 2$ parameters, univariate GARCH for each variance plus a and b

Back

Two-Step Estimators, the Correct Covariance

- Stacked estimating equations: $\sum_t s_{1t}(\theta_1) = 0$ (step 1: GARCH and, with correlation targeting, vech $\bar{Q}$) and $\sum_t s_{2t}(\theta_1, \theta_2) = 0$ (step 2, $\theta_2 = (a, b)$)
- s_{jt}: score of step j on day t; $A_{jk} = \mathbb{E}\left[\partial s_{jt}/\partial \theta_k^\top\right]$, and A_{21} measures how the correlation score reacts to the GARCH parameters
- Newey & McFadden (1994), Section 6: $\sqrt{T}(\hat{\theta}_2 - \theta_2) \to N(0, V_2)$, $V_2 = A_{22}^{-1}\,\mathbb{E}\left[(s_{2t} - A_{21}A_{11}^{-1}s_{1t})(s_{2t} - A_{21}A_{11}^{-1}s_{1t})^\top\right]\,A_{22}^{-1\top}$
- Step-2 standard errors keep only s_{2t}: they drop $A_{21}A_{11}^{-1}s_{1t}$, the GARCH estimation error carried into step 2
- With correct conditional moments the scores are martingale differences and the middle term is their variance; otherwise use its long-run (HAC) version
- Engle & Sheppard (2001) derive V_2 for DCC
- A simple alternative bootstraps the whole two-step procedure (next slide)

Back

Bootstrap of Both DCC Steps

- Parametric residual bootstrap, $B = 199$: decorrelate the residuals with the Cholesky factor of R_t, resample them and simulate DCC innovations with $(\hat{a}, \hat{b}, \bar{Q})$
- Step 2 only: re-estimate (a, b) on the simulated innovations
- Both steps: rebuild returns with the fitted GARCH, re-estimate GARCH, $\bar{Q}$ and (a, b) on the same innovations
- The CCC-null bootstrap of the LR test uses the same machinery with $a = b = 0$: constant-correlation innovations, simulated GARCH returns, both steps re-estimated

Back

Robert Engle and the DCC Model



Robert F. Engle at the Western Economic Association International conference, Santiago, 2017

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- Nobel Memorial Prize in Economic Sciences 2003 for ARCH models (Chapter 5)
- Engle (2002): "Dynamic Conditional Correlation"
  - Keeps the flexibility of univariate GARCH for each variance
  - Adds a simple, estimable law of motion for correlations
- Refinement: the corrected DCC of Aielli (2013), which makes $\bar{Q}$ a consistent target

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Contagion Tests beyond Forbes–Rigobon

- Corsetti, Pericoli & Sbracia (2005): with a common factor the Forbes–Rigobon adjustment over-corrects, biasing the test towards "no contagion"
- They give a factor-model version
- Rigobon (2003): identification through heteroskedasticity
- The variance shift between regimes identifies the structural coefficient β, an instrumental-variables idea
- The common-factor critique applies to the 2008 crisis as well (Seminar A1, B3)

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Copula Densities and Constructions

- Gaussian density: $c = \frac{1}{\sqrt{1-\rho^2}} \exp\left(- \frac{\rho^2(x^2+y^2)-2\rho xy}{2(1-\rho^2)} \right)$, $x = \Phi^{-1}(u)$, $y = \Phi^{-1}(v)$
- t copula: the tail dependence is large for small ν
- Archimedean generators φ in $C(u,v) = \varphi^{-1}\big(\varphi(u) + \varphi(v)\big)$
  - Clayton $\varphi(t) = (t^{-\theta} - 1)/\theta$
  - Gumbel $\varphi(t) = (-\ln t)^{\theta}$
  - Frank $\varphi(t) = -\ln \frac{e^{-\theta t}-1}{e^{-\theta}-1}$
- Asymmetric tails: Clayton for joint crashes of returns, Gumbel for joint extremes of losses
- The 180° rotation $\bar{C}(u,v) = u + v - 1 + C(1-u, 1-v)$ moves the tail to the opposite corner

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Copula Estimates for JPM and BAC, Full Table

Copula	$\hat{\theta}$ or $\hat{\rho}$	$\hat{\nu}$	ℓ	ΔAIC	λ_L	λ_U	fit p
Gaussian	0.833	–	2 486.0	309.3	0.000	0.000	0.005
Student t	0.842	4.24	2 641.7	0.0	0.531	0.531	0.010
Clayton	2.384	–	2 116.7	1 048.0	0.748	0.000	0.005
Gumbel	2.587	–	2 412.2	456.9	0.000	0.693	0.005
Frank	9.152	–	2 390.8	499.8	0.000	0.000	0.005

- CML on pseudo-observations of GARCH residuals
- ΔAIC relative to the best model
- Goodness-of-fit p: Rosenblatt test with parametric bootstrap

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Rank Estimators and Their Precision

- $\hat{\tau}$ is a U-statistic; its Hoeffding projection gives $\sqrt{n}(\hat{\tau} - \tau) \to N\big(0,\ 16\mathrm{Var}[2C(U,V) - U - V]\big)$
- Plug-in: $\widehat{\mathrm{SE}}(\hat{\tau}) = 4\,\mathrm{sd}\big(2\hat{C}_n(\hat{U}_i, \hat{V}_i) - \hat{U}_i - \hat{V}_i\big)/\sqrt{n}$
- Inverting $\hat{\tau}$ gives a robust starting value for the likelihood
- CML sandwich (Genest, Ghoudi & Rivest (1995)): $\sqrt{n}(\hat{\theta} - \theta) \to N(0,\ A^{-1}\Sigma A^{-1})$; A: the Fisher information of the copula
- $\Sigma = \mathrm{Var}\big[\dot{\ell}_\theta(U,V) + W_1(U) + W_2(V)\big]$, $\dot{\ell}_\theta$ the score of $\ln c$
- $W_1(u) = \iint 1\{u \le u'\}\,\partial_u \dot{\ell}_\theta(u', v')\,dC(u', v')$: the correction for estimating the margins by ranks; W_2 likewise in v
- Gaussian copula: variance $(1-\rho^2)^2$ for CML, against $(1-\rho^2)^2/(1+\rho^2)$ with known margins

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Chen and Fan (2006), GARCH Filtering and the Limit

- Model: $r_{it} = \mu_i + \sigma_{it}\varepsilon_{it}$ with GARCH σ_{it}
- Innovations $(\varepsilon_{1t}, \varepsilon_{2t})$ i.i.d. with copula C_θ
- Chen & Fan (2006): CML on the ranks of the *estimated* residuals $\hat{\varepsilon}_{it}$ has the same limit distribution as CML on the ranks of the true ε_{it}
- Monte Carlo: 500 samples, $T = 2\,000$, Gaussian copula $\rho = 0.8$, Student t(5) innovations, GARCH as fitted to JPM and BAC
  - Standard deviation of $\hat{\rho}$: 0.0081 (true innovations) and 0.0082 (GARCH residuals)
  - Mean sandwich SE 0.0081, mean inverse-Hessian SE 0.0063

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Nonparametric Tail Dependence, Conditions

- Schmidt & Stadtmüller (2006): the estimator $\hat{\lambda}_L$ of the main text is consistent and asymptotically Normal if $k \to \infty$, $k/n \to 0$, the tail copula exists and a second-order bias condition holds ($\sqrt{k}\, A(n/k) \to 0$, $A(\cdot)$ the rate at which $C(q,q)/q$ approaches λ_L)
- The i.i.d. bootstrap of residual pairs assumes a static innovation copula
- The stationary block bootstrap keeps the serial dependence of the difference $1\{U \le q,\, V \le q\} - 1\{U > 1-q,\, V > 1-q\}$, whose mean divided by q estimates $\lambda_L - \lambda_U$

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Dependence in High Dimension and over Time

- DCC for large N: the full likelihood is slow and biased
- Remedies: composite likelihood over pairs of assets (Pakel, Shephard, Sheppard & Engle (2021)) and DCC-NL, nonlinear shrinkage of the target $\bar{Q}$ (Engle, Ledoit & Wolf (2019); the shrinkage of Chapter 4)
- Vine (pair-copula) constructions (Aas, Czado, Frigessi & Bakken (2009)): $d(d-1)/2$ bivariate copulas, some conditional, in nested trees, each with its own family and tails
- Hierarchical Archimedean copulas (Okhrin, Okhrin & Schmid (2013)): groups nested by the strength of dependence, e.g. banks within a country and countries within a region
- Dynamic copulas: score-driven parameters (Creal, Koopman & Lucas (2013)) extend to many assets through factor copulas
- Oh & Patton (2018) measure time-varying systemic risk in credit default swap spreads
- Conditional copulas: Patton (2006)

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From Correlations to Systemic Risk

- Correlation networks are the first step towards the systemic-risk measures of Chapter 18
  - Connectedness from Granger causality: Billio, Getmansky, Lo & Pelizzon (2012)
  - Spillover indices from variance decompositions: Diebold & Yilmaz (2009)
  - Connectedness from quantile regressions: CoVaR, Adrian & Brunnermeier (2016)
- Oil price shocks and the US stock market: Kilian & Park (2009)
- Correlation networks inherit all the problems of correlations: non-linearity, time variation, tails

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COVOL: Data and Estimation

- **Model** (Eq. 8–11 of Engle & Campos-Martins (2023)): for $i \neq j$, $\Psi_{ij,t} = s_i s_j v_t$, $v_t = \mathrm{Var}_{t-\mathbf{1}}(x_t)$, so Ψ has one factor and its first principal component gives starting values (Eq. 12–13)
  - e_{it}: the standardised residual of asset i; $\Psi_{ij,t} = \mathrm{Cov}_{t-\mathbf{1}}(e_{it}^{\mathbf{2}}, e_{jt}^{\mathbf{2}})$
  - Equal loadings (before the normalisation $s^\top s = 1$): $g_{it} = x_t$, and the cross-sectional mean $\hat{x}_t = \frac{\mathbf{1}}{N}\sum_i e_{it}^{\mathbf{2}}$ of the main text is the maximum-likelihood estimate (Eq. 17)
- **Estimation** (Section 5.3): maximise $-\frac{\mathbf{1}}{\mathbf{2}}\sum_{i,t}\left[\ln g_{it} + e_{it}^{\mathbf{2}}/g_{it}\right]$, $g_{it} = s_i x_t + 1 - s_i$, alternating x_t and s_i until convergence
  - After each step: $s^\top s = 1$, mean of x_t equal to 1 (Remark 2)
  - Test ξ (Eq. 27): standardised mean of $(e_{it}^{\mathbf{2}} - 1)(e_{jt}^{\mathbf{2}} - 1)$ over the $N(N-1)/2$ pairs, $\mathcal{N}(0,1)$ under $H_{\mathbf{0}}$
- **Data** (Section 7.1, Table B.1): daily USD log returns of 47 country ETFs listed on the NYSE, 2 June 2000 – 1 March 2021
- Same closing time, so no asynchrony
- Unbalanced panel
  - Mean: AR(1) plus two factors, the ACWI and PC1 of the ETF returns
  - GARCH(1,1) variances
  - COVOL fitted to the 47 residual series and the two factors
  - Also: 23 US asset-class series (Section 9) and Monte Carlo studies of the estimator and of the test (Sections 5.4, 6.1)

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COVOL: Key Results

Day	$\hat{x}_t$	$\bar{r}_t$ (%)	Event (label of the paper)
9 March 2020	51.416	−9.650	COVID-19: Italy locked down; oil price war
24 June 2016	49.190	−7.270	Brexit
5 August 2019	39.457	−2.574	Trade war: China devalues its currency
28 April 2020	38.883	2.334	COVID-19: tech share sell-off
17 September 2001	30.077	−8.630	NYSE reopens after 9/11

$\bar{r}_t$: mean return across the country ETFs; top five of the twenty largest values in Table 6 of Engle & Campos-Martins (2023)

- Loadings (Table 7): PC1 0.261 and ACWI 0.250 lead
- France 0.217, Spain 0.200 against New Zealand 0.065, Poland 0.080: factors load more than idiosyncrasies, so correlations rise with COVOL (Eq. 41)
- ACWI squared residual on COVOL_m^2 (monthly mean of $\hat{x}_t$ minus 1): 0.326 (SE 0.025), $R^2 = 0.404$ (Table 8)
- Mean pairwise ETF correlation +0.053 per unit (SE 0.009, Table 10, column 4)
- Change in VIX (Cboe index of the implied volatility of the S&P 500) 4.034 (SE 0.249), $R^2 = 0.517$ (Table 12)

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COVOL: Replication Protocol

- **Replicate** the asset-class version (Section 9) with the US-listed ETFs from the course market data, which all close at the same time
- The ETFs: 11 sector ETFs, Russell 1000 growth and value, Russell 2000, developed and emerging markets outside the US, gold, long Treasuries, corporate bonds, crude oil
- One factor, PC1, on an unbalanced panel
- $\hat{x}_t$ by Eq. 17, then the iteration of Section 5.3
- The ξ test before and after dividing e_{it}^2 by $\hat{g}_{it}$ (Section 7.2)
- Table 15 of the paper ranks 9 November 2016 (43.04) and 9 March 2020 (40.47) among the top four days
- The dollar and the Russell 2000 growth/value split are not in the course data

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COVOL: Limitations and the Data after 2021

- Limitations stated by the authors: after dividing by $\hat{g}_{it}$ the squares are still correlated, possibly a second factor (Section 7.2); multifactor COVOL is open (Section 9)
- With $s^\top s = 1$ the loadings shrink as N grows: R_s^2 (the fit between estimated and true loadings) is 0.640 for $N = 50$, $T = 1000$, against 0.924 for $N = 10$ (Table 2)
- The test loses power with unequal loadings, so the authors advise $T \geq 1000$
- After the sample of the paper: 3 April 2025 ($\hat{x}_t = 16.2$, mean return -3.67%), after the US tariff announcement; 10 November 2022 ($\hat{x}_t = 15.4$, $+4.56\%$), a lower-than-expected US inflation print
- Open question (Seminar C3): do elections, tariffs and inflation news share one factor?
- Open question: does COVOL_m^2 explain the monthly DCC correlations beyond the change in VIX?

The Discovery Loop, Reusable Prompts (1/2)

- Tools are interchangeable: an AI assistant (Claude, ChatGPT, Gemini, GitHub Copilot) and literature engines (Semantic Scholar, Elicit, Google Scholar)
- **Literature**: *"List studies since 2015 on the drivers of the stock–bond correlation, with DOI, sample, measure of correlation, identification strategy and main estimate."*
  - Resolve every DOI on Crossref and check the estimate in the paper, not in the summary
- **Hypotheses**: *"Give competing explanations for the 2022 sign change (inflation level, monetary-policy regime, term-premium shock, a one-off tightening) and what each predicts for ρ before 2022 and out of sample."*
- **Data and code**: *"Write Python that aligns SPY and TLT adjusted prices from the course market data on common trading days, computes monthly realised correlations and merges US CPI inflation published before each month starts."*
  - Replicate first: -0.40 (2002–2021) and 0.11 (2022–2026)

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The Discovery Loop, Reusable Prompts (2/2)

- **Robustness**: *"Write a pre-registered plan: the pre-2022 subsample, core CPI instead of headline, weekly instead of daily returns, DCC instead of realised correlation, control for realised volatility, HAC lag 12."*
  - Placebo: replace π by simulated random-walk series of the same persistence
  - A persistent regressor can produce large t-statistics by chance
- **Critique**: *"Act as a referee: is γ_1 identified by a single episode, and would the Forbes–Rigobon volatility bias explain it? Give the test that would expose each."*
- **Report**: every inflation measure, window and frequency tried
- Subsample estimates next to the full sample
- Any specification chosen after seeing results is exploratory and reported as such
- Code of the mini-case: Quantlets/Ch_06/ai_discovery_case.py
- Inflation: US consumer price index for all urban consumers, U.S. Bureau of Labor Statistics

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Acronyms used in this material (1/4)

- **ACWI**: All Country World Index (MSCI)
- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AR**: AutoRegressive (model)
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **BAC**: Bank of America Corporation (NYSE ticker)
- **BEKK**: Baba–Engle–Kraft–Kroner (multivariate GARCH model)
- **BET**: Bucharest Exchange Trading (index)
- **BIC**: Bayesian Information Criterion
- **BNP**: BNP Paribas SA (Euronext Paris ticker; Banque Nationale de Paris)
- **BRD**: BRD–Groupe Société Générale (BVB ticker)
- **CCC**: Constant Conditional Correlation (model)

Acronyms used in this material (2/4)

- **CDO**: Collateralised Debt Obligation
- **CML**: Canonical Maximum Likelihood
- **COVID-19**: COronaVIrus Disease 2019
- **COVOL**: COmmon VOLatility (global COVOL: the size of the volatility shock common to all assets)
- **CPI**: Consumer Price Index
- **DBK**: Deutsche Bank AG (Xetra ticker)
- **DCC**: Dynamic Conditional Correlation
- **DCC-NL**: Dynamic Conditional Correlation with NonLinear shrinkage
- **DOI**: Digital Object Identifier
- **EPU**: Economic Policy Uncertainty (index)
- **ES**: Expected Shortfall
- **ETF**: Exchange-Traded Fund

Acronyms used in this material (3/4)

- **EU**: European Union
- **EWMA**: Exponentially Weighted Moving Average
- **FRM**: Financial Risk Meter
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GAS**: Generalised Autoregressive Score (model)
- **GPR**: GeoPolitical Risk (index)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **i.i.d.**: independent and identically distributed
- **IFM**: Inference Functions for Margins
- **JPM**: JPMorgan Chase & Co. (NYSE ticker)
- **LR**: Likelihood Ratio (test)
- **MSCI**: Morgan Stanley Capital International (index provider)

Acronyms used in this material (4/4)

- **NYSE**: New York Stock Exchange
- **PC1**: first Principal Component
- **SE**: Standard Error
- **SPY**: SPDR S&P 500 ETF Trust
- **TLT**: iShares 20+ Year Treasury Bond ETF
- **TLV**: Banca Transilvania (BVB ticker)
- **US**: United States
- **USD**: United States Dollar
- **VEC**: VECtorised multivariate GARCH model (vech of the covariance matrix)
- **VIX**: Cboe Volatility Index (implied volatility of the S&P 500)

References (1/5)

- Aas, K., Czado, C., Frigessi, A., Bakken, H. (2009). Pair-copula constructions of multiple dependence. *Insurance: Mathematics and Economics*, 44(2), 182–198.
- Adrian, T., Brunnermeier, M.K. (2016). CoVaR. *American Economic Review*, 106(7), 1705–1741.
- Aielli, G.P. (2013). Dynamic conditional correlation: on properties and estimation. *Journal of Business & Economic Statistics*, 31(3), 282–299.
- Ang, A., Chen, J. (2002). Asymmetric correlations of equity portfolios. *Journal of Financial Economics*, 63(3), 443–494.
- Baker, S.R., Bloom, N., Davis, S.J. (2016). Measuring economic policy uncertainty. *Quarterly Journal of Economics*, 131(4), 1593–1636.
- Billio, M., Getmansky, M., Lo, A.W., Pelizzon, L. (2012). Econometric measures of connectedness and systemic risk in the finance and insurance sectors. *Journal of Financial Economics*, 104(3), 535–559.
- Bollerslev, T. (1990). Modelling the coherence in short-run nominal exchange rates: a multivariate generalized ARCH model. *Review of Economics and Statistics*, 72(3), 498–505.
- Bollerslev, T., Engle, R.F., Wooldridge, J.M. (1988). A capital asset pricing model with time-varying covariances. *Journal of Political Economy*, 96(1), 116–131.
- Caldara, D., Iacoviello, M. (2022). Measuring geopolitical risk. *American Economic Review*, 112(4), 1194–1225.
- Campbell, J.Y., Pflueger, C., Viceira, L.M. (2020). Macroeconomic drivers of bond and equity risks. *Journal of Political Economy*, 128(8), 3148–3185.
- Chen, X., Fan, Y. (2006). Estimation and model selection of semiparametric copula-based multivariate dynamic models under copula misspecification. *Journal of Econometrics*, 135(1–2), 125–154.
- Clayton, D.G. (1978). A model for association in bivariate life tables and its application in epidemiological studies of familial tendency in chronic disease incidence. *Biometrika*, 65(1), 141–151.
- Corsetti, G., Pericoli, M., Sbracia, M. (2005). 'Some contagion, some interdependence': more pitfalls in tests of financial contagion. *Journal of International Money and Finance*, 24(8), 1177–1199.

References (2/5)

- Creal, D., Koopman, S.J., Lucas, A. (2013). Generalized autoregressive score models with applications. *Journal of Applied Econometrics*, 28(5), 777–795.
- Demarta, S., McNeil, A.J. (2005). The t copula and related copulas. *International Statistical Review*, 73(1), 111–129.
- Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- Diebold, F.X., Yilmaz, K. (2009). Measuring financial asset return and volatility spillovers, with application to global equity markets. *Economic Journal*, 119(534), 158–171.
- Embrechts, P., McNeil, A.J., Straumann, D. (2002). Correlation and dependence in risk management: properties and pitfalls. In M.A.H. Dempster (ed.), *Risk Management: Value at Risk and Beyond*, 176–223. Cambridge University Press.
- Engle, R.F. (2002). Dynamic conditional correlation: a simple class of multivariate generalized autoregressive conditional heteroskedasticity models. *Journal of Business & Economic Statistics*, 20(3), 339–350.
- Engle, R.F., Campos-Martins, S. (2023). What are the events that shake our world? Measuring and hedging global COVOL. *Journal of Financial Economics*, 147(1), 221–242.
- Engle, R.F., Kroner, K.F. (1995). Multivariate simultaneous generalized ARCH. *Econometric Theory*, 11(1), 122–150.
- Engle, R.F., Ledoit, O., Wolf, M. (2019). Large dynamic covariance matrices. *Journal of Business & Economic Statistics*, 37(2), 363–375.
- Engle, R.F., Sheppard, K. (2001). Theoretical and empirical properties of dynamic conditional correlation multivariate GARCH. NBER Working Paper 8554.
- Forbes, K.J., Rigobon, R. (2002). No contagion, only interdependence: measuring stock market comovements. *Journal of Finance*, 57(5), 2223–2261.
- Frank, M.J. (1979). On the simultaneous associativity of F(x, y) and x + y − F(x, y). *Aequationes Mathematicae*, 19, 194–226.

References (3/5)

- Franke, J., Härdle, W.K., Hafner, C.M. (2019). Statistics of Financial Markets: An Introduction, 5th ed. Springer.
- Genest, C., Ghoudi, K., Rivest, L.-P. (1995). A semiparametric estimation procedure of dependence parameters in multivariate families of distributions. *Biometrika*, 82(3), 543–552.
- Genest, C., Rémillard, B., Beaudoin, D. (2009). Goodness-of-fit tests for copulas: a review and a power study. *Insurance: Mathematics and Economics*, 44(2), 199–213.
- Genest, C., Rivest, L.-P. (1993). Statistical inference procedures for bivariate Archimedean copulas. *Journal of the American Statistical Association*, 88(423), 1034–1043.
- Gumbel, E.J. (1960). Bivariate exponential distributions. *Journal of the American Statistical Association*, 55(292), 698–707.
- Härdle, W.K., Simar, L. (2019). Applied Multivariate Statistical Analysis, 5th ed. Springer.
- Herskovic, B., Kelly, B., Lustig, H., Van Nieuwerburgh, S. (2016). The common factor in idiosyncratic volatility: quantitative asset pricing implications. *Journal of Financial Economics*, 119(2), 249–283.
- Joe, H. (2005). Asymptotic efficiency of the two-stage estimation method for copula-based models. *Journal of Multivariate Analysis*, 94(2), 401–419.
- J.P. Morgan/Reuters (1996). RiskMetrics – Technical Document, 4th ed. New York.
- Kendall, M.G. (1938). A new measure of rank correlation. *Biometrika*, 30(1–2), 81–93.
- Kilian, L., Park, C. (2009). The impact of oil price shocks on the U.S. stock market. *International Economic Review*, 50(4), 1267–1287.
- Li, D.X. (2000). On default correlation: a copula function approach. *Journal of Fixed Income*, 9(4), 43–54.
- Liu, Y., Tsyvinski, A. (2021). Risks and returns of cryptocurrency. *Review of Financial Studies*, 34(6), 2689–2727.

References (4/5)

- Longin, F., Solnik, B. (2001). Extreme correlation of international equity markets. *Journal of Finance*, 56(2), 649–676.
- Mantegna, R.N. (1999). Hierarchical structure in financial markets. *European Physical Journal B*, 11(1), 193–197.
- Markowitz, H. (1952). Portfolio selection. *Journal of Finance*, 7(1), 77–91.
- Molenaar, R., Sénéchal, E., Swinkels, L., Wang, Z. (2024). Empirical evidence on the stock–bond correlation. *Financial Analysts Journal*, 80(3), 17–36.
- Nelsen, R.B. (2006). An Introduction to Copulas, 2nd ed. Springer.
- Newey, W.K., McFadden, D. (1994). Large sample estimation and hypothesis testing. In R.F. Engle, D.L. McFadden (eds.), *Handbook of Econometrics*, Vol. 4, 2111–2245. Elsevier.
- Newey, W.K., West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- Oh, D.H., Patton, A.J. (2018). Time-varying systemic risk: evidence from a dynamic copula model of CDS spreads. *Journal of Business & Economic Statistics*, 36(2), 181–195.
- Okhrin, O., Okhrin, Y., Schmid, W. (2013). On the structure and estimation of hierarchical Archimedean copulas. *Journal of Econometrics*, 173(2), 189–204.
- Pakel, C., Shephard, N., Sheppard, K., Engle, R.F. (2021). Fitting vast dimensional time-varying covariance models. *Journal of Business & Economic Statistics*, 39(3), 652–668.
- Patton, A.J. (2006). Modelling asymmetric exchange rate dependence. *International Economic Review*, 47(2), 527–556.

References (5/5)

- Politis, D.N., White, H. (2004). Automatic block-length selection for the dependent bootstrap. *Econometric Reviews*, 23(1), 53–70.
- Rémillard, B. (2017). Goodness-of-fit tests for copulas of multivariate time series. *Econometrics*, 5(1), 13.
- Rigobon, R. (2003). Identification through heteroskedasticity. *Review of Economics and Statistics*, 85(4), 777–792.
- Schmidt, R., Stadtmüller, U. (2006). Non-parametric estimation of tail dependence. *Scandinavian Journal of Statistics*, 33(2), 307–335.
- Sklar, A. (1959). Fonctions de répartition à n dimensions et leurs marges. *Publications de l'Institut de Statistique de l'Université de Paris*, 8, 229–231.
- Spearman, C. (1904). The proof and measurement of association between two things. *American Journal of Psychology*, 15(1), 72–101.
- Tse, Y.K. (2000). A test for constant correlations in a multivariate GARCH model. *Journal of Econometrics*, 98(1), 107–127.
- Wied, D., Krämer, W., Dehling, H. (2012). Testing for a change in correlation at an unknown point in time using an extended functional delta method. *Econometric Theory*, 28(3), 570–589.